Temperature and Local Industry Concentration

by

Jacopo Ponticelli
Northwestern University

Qiping Xu
UIUC

Stefan Zeume
UIUC

CES 23-51 **October 2023**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Christopher Goetz, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

We use plant-level data from the US Census of Manufacturers to study the short and long run effects of temperature on manufacturing activity. We document that temperature shocks significantly increase energy costs and lower the productivity of small manufacturing plants, while large plants are mostly unaffected. In US counties that experienced higher increases in average temperatures between the 1980s and the 2010s, these heterogeneous effects have led to higher concentration of manufacturing activity within large plants, and a reallocation of labor from small to large manufacturing establishments. We offer a preliminary discussion of potential mechanisms explaining why large manufacturing firms might be better equipped for long-run adaptation to climate change, including their ability to hedge across locations, easier access to finance, and higher managerial skills.

Keyword*:* Climate Change, Manufacturing, US, Local Industry Concentration

JEL Classification: Q54, O14, G3, L11

[*] This draft benefited from valuable comments from Sophie Shive (discussant), Viral Acharya, Tatyana Deryugina, David Molitor, Juan-Manuel Castro Vincenzi, and Karthik Sastry. Jora Li provided outstand ing research assistance. Ponticelli gratefully acknowledges financial support from the Financial Institu tions and Markets Research Center at Northwestern Kellogg School of Management. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Re view Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 1850. (CBDRB-FY23-P1850-R10208).

Ponticelli: Kellogg School of Management, Northewestern University, NBER and CEPR. Email: jacopo.ponticelli@kellogg.northwestern.edu.
Xu: Gies College of Business, University of Illinois Urbana Champaign. Email: qipingxu@illinois.edu.
Zeume: Gies College of Business, University of Illinois Urbana Champaign. Email: zeume@illinois.edu.

I Introduction

Average temperatures in the continental United States have increased substantially over the past century (IPCC, 2021). The pace of warming has accelerated starting in the 1980s, with a median increase in temperature across US counties of 0.6°C, and nine in ten counties experiencing an increase in their average temperatures between the 1980s decade and the 2010s decade. Understanding the impact of a warming climate[1] on economic activity has gained the center of the political and academic debate, a debate spurred by countless scientific reports and newspaper articles.[2]

We contribute to this debate by providing new evidence on the impact of temperature variation on U.S. manufacturing activity using four decades of plant-level data from the U.S. Census Bureau. Three features of the Census data are particularly relevant. First, the availability of detailed establishment-level characteristics, such as energy costs and productivity, allows a comprehensive anatomy of the impact of temperature shocks. Second, the ability to observe the cross-section of plants allows us to study the heterogeneous effects of temperature shocks on establishments of different sizes. Third, observing plant performance over four decades enables us to study the response of manufacturing activity to long-run temperature changes. To the extent that plants differ in their sensitivity to temperature variation, one may expect manufacturing activity to concentrate among those firms that are better able to adapt. Indeed, the main contribution of this paper is to document the impact of temperature variation on local industry concentration, and to discuss its potential drivers.

Our empirical strategy builds on two approaches to estimate the effect of temperature on manufacturing plants. The first approach aims at capturing the contemporaneous response of manufacturing outcomes to short-term (yearly) temperature shocks. The second approach aims at capturing the long-term response of manufacturing activity to changes in the average climate experienced by a county in the last four decades. We combine the Census of Manufacturing Firms (CMF) and the Annual Survey of Manufacturers (ASM) to measure plants' energy costs, productivity, and size. We use the Longitudinal Business Database (LBD), an administrative register that tracks all business establishments in the US, to identify plant entry and exit in different geographic locations.

We start by estimating a panel regression at the plant-year level which exploits yearly variation in temperature in the ZIP code where the plant is located. We think of these

[1] As is common practice, we use the term weather to refer to realizations of temperature, drawn from an underlying distribution, and climate to refer to moments of the weather distribution (e.g., Auffhammer (2018), Dell et al. (2012)).

[2] See, for instance, reports from the following agencies and organizations:
US Environmental Protection Agency: www.epa.gov/climateimpacts/climate-change-impacts-sector
World Health Organization: https://www.who.int/publications/i/item/9789240057906
International Labour Organization: https://www.ilo.org/global/publications/books/WCMS_711919
White House: https://www.whitehouse.gov/briefing-room/statements-releases/2021/09/20

Electronic copy available at: https://ssrn.com/abstract=4526111

yearly temperature shocks as random "weather" draws from the "climate" distribution in a given geographical area, and therefore as plausibly exogenous to the outcomes of interest (Dell et al., 2014). Because of our focus on manufacturing, we think of each US ZIP code or county as a small open economy and manufacturing as a tradable sector whose demand is geographically sparse across the US and the rest of the world, and thus relatively independent from local demand shocks. Under this assumption, temperature shocks are likely to identify supply forces such as higher input costs or negative labor productivity shocks, rather than any effect of temperature on local demand of the goods produced by each plant.

Two key findings emerge from our estimates of the short-run effects of temperature shocks on manufacturing outcomes. First, input costs associated with temperature management – mainly expenditures in electricity and fuel – and productivity react to contemporaneous temperature shocks. In particular, warmer than usual temperatures increase energy spending and decrease plant productivity. Second, these effects are concentrated in small manufacturing plants, while large establishments are mostly unaffected. Despite these contemporaneous negative effects on small plants, we observe no significant contemporaneous response of small plants via down-scaling (as measured by employment) or via exiting a given location, although hotter than usual temperatures dissuade entry. Indeed, it is plausible that key industrial decisions such as scaling back on the size of an existing plant or exiting a given market are not driven by yearly weather shocks, especially if such shocks are interpreted as idiosyncratic and therefore likely to revert in the following years.

In the long-run, however, the cumulative effect of several years of warmer than usual weather might result in corporate responses on the intensive and extensive margins. This is because a series of deviations from past temperatures might indicate a shift in the climate distribution from which weather events are drawn in a given geographical area.

In order to test the idea that plants respond to long-run changes in climate, we use a long differences approach as in Burke and Emerick (2016). In particular, we estimate a county-level regression relating long-run changes in manufacturing activity to long-run changes in average temperatures between the 1980s decade and the 2010s decade, controlling for state-specific common trends and for differential trends across counties with different initial observable characteristics. We find that, over the last four decades, areas that got warmer at a faster pace experienced larger declines in the number of small plants but no differential change in the number of large plants. While the number of large plants did not increase, our results on employment indicate that large plants were able to absorb at least part of the labor force lost by smaller manufacturing establishments. This is consistent with a more pronounced reallocation of employment from small to large plants in counties that experienced a higher increase in temperature in the last four decades.

Based on the heterogeneous responses by small and large plants documented so far,

Electronic copy available at: https://ssrn.com/abstract=4526111

we next investigate the impact of higher average temperatures on different measures of industrial concentration at the county level. We find that counties that in the 2010s decade had a standard deviation higher increase in temperature – about 100 cooling degree days (CDDs) per year above 18°C – relative to the 1980s decade, experienced a 1.4 percentage points larger increase in the share of employment concentrated in the top 5 largest plants, and a 5.1 percent larger increase in the Herfindahl-Hirschman concentration index.[3] Overall, the results indicate that faster warming in the last four decades has led to higher concentration of industrial activity among larger plants. Consistent with these results, we also document a large and significant increase in average size of manufacturing plants in counties that are getting warmer at a faster pace (7.7% larger average plant size for a one-standard-deviation increase in CDDs).

The finding that faster warming has led to higher concentration of manufacturing activity among larger firms suggests that such firms might be better equipped for long-run adaptation to climate change. We conclude the paper by discussing several potential mechanisms that can rationalize this result. First, large firms might be naturally better hedged to absorb weather shocks, even when they occur at higher frequency due to climate change, because they produce output in different locations, diversifying climate risk (Castro-Vincenzi, 2022). Second, large firms might also have better access to external finance, which allows them to cope with weather shocks, reducing the need to downscale employment or close plants. Finally, large firms might have better trained managers who can both understand the change in firm exposure to climate risk and proactively invest in adaptation.

Related Literature

Climate economists have provided ample evidence on the relation between climate change and macroeconomic outcomes (see Dell et al. (2014) for a comprehensive list of outcomes studied and methods employed in the literature). Of particular interest to us is the work on country-level output and productivity. That work generally documents adverse effects of climate change in developing economies and explains the general lack of effects in developed countries by adaptation (Burke et al. 2015, Chen and Yang 2019, Colacito et al. 2019, Dell et al. 2009, Dell et al. 2012, Gallup et al. 1999, Hsiang 2010, Jones and Olken 2010).[4]

[3] A daily Cooling Degree Day (CDD) is the difference in degrees between average daily temperature and 18°C conditional on the average daily temperature being above 18°C, see Heutel et al. (2021) or Zivin and Kahn (2016). Yearly CDD is the sum of all daily CDDs in a given year. A county with average temperature of 23°C on 20 days of the year and average temperature below 18°C the rest of the year will have a CDD of 100.

[4] A notable exception exists in the agricultural sector where short-term temperature shocks are generally found to have adverse implications for productivity in developed nations once nonlinearities are considered. See, for instance, Burke and Emerick (2016), Fisher et al. (2012), Ortiz-Bobea et al. (2018), Schlenker and Roberts (2009) for evidence on the US, Lobell et al. (2011) for global evidence, Gupta et al. (2017) and Auffhammer et al. (2006) for evidence on India. In addition, Deryugina and Hsiang

Electronic copy available at: https://ssrn.com/abstract=4526111

Within this area of inquiry, we focus on the impact of temperature shocks and long-run changes in temperature on establishments in the U.S. manufacturing sector. Though in developed economies a higher capacity of adaptation may be presumed the norm, in manufacturing not all facilities are climate-controlled, the production process itself produces substantial heat, and the power grid providing energy to industrial establishments is subject to stress when temperatures rise above certain thresholds (Graff Zivin and Neidell, 2014). Moreover, heterogeneity in establishment size and organizational structure may affect establishments' ability to respond to as well as their incentives to prepare for temperature shocks (Castro-Vincenzi 2022, Zivin and Kahn 2016, Somanathan et al. 2021). Consistent with the idea that firms are heterogeneous with respect to their ability to absorb temperature shocks and adapt to climate change, we find temperature shocks and climate change to have adverse average effects on U.S. manufacturing firms. These effects are almost entirely concentrated among small firms, and they have implications for local industry concentration.

More specifically, our findings relate contribute to the rapidly growing body of empirical work on climate and firms. Among others, higher temperatures have been documented to reduce sales among non-U.S. global firms (Pankratz et al. 2023; using the 1995-2019 period), affect the profitability of public firms across more than 40 percent of U.S. industries (Addoum et al. 2021; 1990-2015), lead to higher yield for bonds and conditional returns for equity for US firms (Acharya et al. 2022), and reduce total factor productivity for Chinese manufacturing firms (Zhang et al. 2018; 1998-2007).

Existing work has also documented that clients dynamically adjust their supplier network in response to weather shocks (Custodio et al. 2022, Pankratz and Schiller 2021; both using the 2000-2015 period). In particular, suppliers of the same client lose two percent in sales when faced with a one degree Celsius temperature increase vis-a-vis other same-client suppliers. Contrary to the by-and-large adverse effects on firms documented in these papers, Addoum et al. (2020) find no effects of higher temperatures on establishment sales and productivity of establishments owned by public U.S. firms, which are likely larger plants.[5] In order to speak to this apparent ambiguity in findings, we study the effects of temperature changes on a representative sample of U.S. manufacturing establishments, including small- and medium-sized establishments, over four decades. We focus on entry and exit as possible margins of adjustments, and, by including small establishments, are able to highlight that small firms bear the costs of a warming climate. Contrary to existing work on firms, our sample period covers over four decades (1977-2018), and is hence not dominated by the so-called hiatus, a 16-year period (1998-2013) of slow average warming (Cahill et al. (2015), Hsiang and Kopp (2018)). Length of the

(2014) show that temperature affects income even in the US.

[5]Other related work has used different climate shocks, such as floods, to document effects on entry, employment, and output, see, e.g., Desmet and Rossi-Hansberg (2021), Lin et al. (2021), and Jia et al. (2022).

Electronic copy available at: https://ssrn.com/abstract=4526111

sample period and granularity of the data allow us to include multiple high-order fixed effects all the while considering non-linearities along the full spectrum of the temperature distribution.

Our results further speak directly to the notion that productive firms, which also contribute more to overall industry productivity, have greater incentives to adapt (Zivin and Kahn (2016), Somanathan et al. (2021)). With this, we think our findings are informative also for the literature on adaptation, i.e., the speed of adjustment to changes in the environment. While Samuelson (1947) and Viner (1958) provide theoretical guidance, empirical evidence has been mixed and setting-dependent. In agriculture, for instance, Burke and Emerick (2016) find long-run adaptation to offset less than half and possibly none of the short-term losses inflicted by increasing temperatures on corn yields. Similarly, Hornbeck (2012) finds effects of soil erosion during the 1930s to have effects on productivity even two decades later, though Olmstead and Rhode (2011) find some evidence of adaptation in wheat cultivation to harsher environments.[6] One means of adaptation that has received considerable attention in climate studies and constitutes a likely force behind our results is the availability of air conditioning (AC). Somanathan et al. (2021) show that climate control mitigates some of the productivity losses caused by heat but also point out that in some industries, such as the garment industry, adaptation of AC may not be justified given that the additional electricity costs more than offset value-added. This point–that more productive firms have greater incentives to adapt–is formalized in Zivin and Kahn (2016) and exploited to show that the overall impact of heat on industry productivity is dampened: productive firms have a higher output share and are more likely to adopt AC. Adaptation to floods, in the meantime, is front-and-center in Castro-Vincenzi (2023) where large car manufacturers reallocate production away from affected plants. By-and-large, our evidence on small firms and the mediating role played by energy costs corroborates these findings. More broadly, while the adverse effects of climate change on firms are for the most part found among developing economies (Dell et al. (2012), Burke et al. (2015)), we show that they matter also in U.S. manufacturing, especially to small firms.

Finally, our findings are linked closely to the literature on industry concentration, a literature that documents increasing trends in industry concentration for the U.S. over recent decades (e.g., De Loecker et al. (2020), Grullon et al. (2019)). Drivers behind this increased industry concentration are broadly of technological or political nature, with work focusing on channels such as the efficient scale of operation (Autor et al. (2017), Autor et al. (2020)), the decrease in domestic competition (Gutiérrez and Philippon (2017)), and the increasing importance of globalization (Feenstra and Weinstein (2017)), as well as the

[6]To illustrate the range of settings to which theories of adaptation have been applied, Davis and Weinstein (2002) and Miguel and Roland (2006) show temporary negative shocks to have no adverse long-lasting effects on economic and quantitative demographic outcomes in a very different setting, that of the bombings of Japan and Vietnam.

Electronic copy available at: https://ssrn.com/abstract=4526111

shift away from physical to intangible capital (Alexander and Eberly (2018), Crouzet and Eberly (2021)). To this debate, we add climate change as a driver which, through its adverse impact on small firms, contributes to increased local industry concentration.

The rest of the paper is organized as follows. Section II describes the data and offers some background information on changes in average temperatures experienced in the continental U.S. in recent decades. Section III presents the identification strategy. Section IV discusses the short-run and long-run effects of temperature changes on manufacturing activity. Section V offers a preliminary discussion of mechanisms that can rationalize our findings.

II Data and Background

II.A Data

Manufacturing establishments data. To measure manufacturing activity, we rely on three complementary establishment-level data sets from the U.S. Census Bureau. First, we employ the Longitudinal Business Database (LBD), an administrative register that tracks all business establishments. LBD provides information on establishment geographic locations and industry classification. We further exploit data on the number of employees to distinguish by establishment size. We have access to the LBD data for the time period of 1977-2019.

Second, we combine the Census of Manufacturing Firms (CMF) and the Annual Survey of Manufacturers (ASM) for details on activities of manufacturing establishments, which are classified using 2-digit NAICS code 31-33. The CMF covers all US manufacturing plants with at least one employee and it is carried out every five years. The ASM provides data on non-Census years for a sample of 50,000 to 70,000 manufacturing establishments, including all establishments with more than 250 employees and a representative sample of smaller establishments. Sampling weights are included for all plant-years to reflect that smaller plants are less likely to be surveyed relative to their large peers.[7] We construct a panel at the plant-year level in which we use plants covered by the ASM in non-Census years, and CMF plants that are also observed in the ASM in Census years. The ASM/CMF data span the time period 1973-2018. From these two data sets, we use detailed industry classification, identification of business group affiliation, output (measured by value of shipments), energy costs, total working hours, and employment. We also use total factor productivity (TFP) as in Foster et al. (2016) (see their Appendix). Data quality is ensured through a mandatory reporting requirement and fines for misreporting.

[7] See Foster et al. (2016) and Ersahin et al. (2021) for further detail.

Electronic copy available at: https://ssrn.com/abstract=4526111

Weather data. We use two data sources to capture the weather and temperature-related changes in climate, as well as other climate shocks, respectively. Weather data for the contiguous United States over the 1950-2019 period is provided by the PRISM Climate Group. To suit our purpose, we rely on the cleaned version provided on Wolfram Schlenker's homepage.[8] Data includes daily minimum and maximum temperatures for 2.5 by 2.5 mile grids on the basis of a constant set of weather stations that receive a constant weight over the 1950-2019 sample period. This treatment ensures that the resulting time series of temperatures does not vary through the birth and death of stations or missing observations (see Auffhammer et al. (2013) for a discussion of these and other potential pitfalls).

For our plant-level analysis, we measure plants' temperature exposure at the zip code level. In order to obtain zip-code level average daily temperatures, we first calculate the daily average temperature for each 2.5 by 2.5 mile grid as the equally weighted average of daily minimum temperature and daily maximum temperature reported for that grid. For each zip code, we then calculate the value-weighted daily average temperature using grid points within a 20-mile radius of the zipcode centroid and their inverse distance to the centroid as the weight, a method that follows Heutel et al. (2021). For our county-level analysis, we weight all grid-level temperature observations within a county by their inverse distance to the geographic county midpoint. We obtain precipitation information using the same method. On the basis of the resulting respective zip code-day and county-day temperature time series, we construct various aggregate yearly temperature measures of interest, such as number of days within certain temperature bins as well as Cooling Degree Days (CDD) and Heating Degree Days (HDD).

We further obtain data on extreme weather events such as droughts and floods, heatwaves and winter weather, as well as hurricanes and tornadoes from SHELDUS.[9] SHELDUS covers the 1960-2021 period and assigns events to counties; underlying data is from the National Center for Environmental Information and SHELDUS has significantly more records of natural disaster events than alternative data provided by alternative data sources such as the Federal Emergency Management Agency. We use hazards reported by SHELDUS as controls and also to validate our temperature data.

Economic and demographic controls. Three county-level economic and demographic controls are based on the 1980 Census and serve as controls for pre-sample period conditions. Income per capita and population are obtained directly from the Census webpage, and fraction of the population above 25 years of age with a college degree is imputed from

[8]For details such as treatment of missing values and selection of underlying stations, please refer to: `http://www.columbia.edu/~ws2162/links.html`.

[9]ASU Center for Emergency Management and Homeland Security (2023). The Spatial Hazard Events and Losses Database for the United States, Version 21.0 [Online Database]. Phoenix, AZ: Arizona State University. Available from `https://cemhs.asu.edu/sheldus`

Electronic copy available at: https://ssrn.com/abstract=4526111

data provided through IPUMS-NGHIS (the National Historical Geographic Information System). A further control captures the change in exposure to import competition from China over the 1990 to 2007 period and reflects exposure per worker as in Autor et al. (2013) on the basis of UN Comtrade data.

II.B Background on changes in temperature in the US

The contiguous U.S. has experienced substantial increases in average temperature over the last century. According to the climatology literature described in the IPCC (2021) report, the significant emergence of changes in temperature relative to historical averages occurred in North America after 1981.[10] Figure I supports this observation by showing the dynamics of annual average surface temperature anomalies across the contiguous 48 states over the last 120 years. A temperature anomaly is the difference between the average annual temperature and the average temperature over the 1901-2000 period. Figure I shows that after mild increases in average temperature in the 1930s and 1940s, the 1960s and 1970s witnessed a cooling period. In stark contrast, and in line with the IPCC (2021) report, average temperature increased rapidly and consistently after 1980.[11] This trend is particularly pronounced in the 2000s and 2010s, and the 2012 to 2016 period experienced some of the highest abnormal temperatures.

This recent trend of increasing temperatures is predicted to continue for the next decades, as confirmed by long-run projections of U.S. temperatures for the remainder of the 21st century. In Figure II we utilize the "big data" generated by Hsiang et al. (2017) to illustrate predicted long-run trends. These data contain binned projections of daily weather (1981-2100) for US counties using 44 different climate models, and records the number of days that fall within 1-Celsius degree bins within a year (from -20°C to 40°C).[12] Next, we take the average days across all climate models for each county-year, and then calculate the mean value across all counties in a decade.

Climate modeling generally considers four Representative Concentration Pathways (RCPs) to describe different 21st-century pathways of greenhouse gas (GHG) emissions and atmospheric concentrations. The RCPs include a stringent mitigation scenario (RCP2.6), two intermediate scenarios (RCP4.5 and RCP6.0), and one scenario with very high GHG emissions (RCP8.5, frequently referred to as "business as usual" or "worst case scenario"). The most pronounced pattern in Figure II is the sharp spikes in the number of

[10] See IPCC (2021), page 133. Historical climate is calculated using temperature data for the baseline period 1850-1900.

[11] To further illustrate the recent increase in average temperatures, our own analysis shows that over the last four decades, the median increase in average temperature across counties was 0.6°C, an increase in average temperature was observed in more than 90% of all counties, and the average county experienced 3.4 more days per year above 26°C (and 2 fewer days per year below -6°C).

[12] In order to align the arguments in this section with our later analysis, we group temperature projections into coarser bins of 3 Celsius degrees. In particular, we create 11 bins of 3°C each, ranging from -6°C to 26°C, plus two additional bins capturing average daily temperatures below -6°C and above 26°C.

Electronic copy available at: https://ssrn.com/abstract=4526111

extremely hot days, namely days with an average temperature above 26°C. The average number of days above 26°C increases from 20 days in the 2010s to 40 days by the end of the century under the optimistic scenario (RCP2.6), 60 days under the intermediate scenario (RCP4.5), and about 100 days under the worst-case scenario (RCP8.5). In other words, the number of extremely hot days is expected to double under the best-case scenario, and quintuple in the worst-case scenario.

Figure III illustrates the geographic distribution of projected changes in extremely hot days between the 1980s and the 2090s. Across all three RCPs, we observe a prevalent increase in the number of extremely hot days across the US, with the largest increases in hot days predicted to occur in the central and south US counties. Notably, there is also significant variation in projected hot days across counties within each state.

III Empirical Strategy

Our empirical strategy builds on two approaches to estimate the effect of temperature on manufacturing outcomes. The first approach aims at capturing the contemporaneous response of manufacturing outcomes to short-term (yearly) temperature shocks. The second approach aims at capturing the long-term response of manufacturing activity to changes in the average temperature experienced by a county in the last four decades.[13] We describe each approach in turn.

III.A Panel approach to study the short-run effects of temperature

We start by studying the short-run effects of temperature on manufacturing outcomes by estimating the following panel specification at the plant-year level:

$$y_{ijz(s)t} = \alpha_i + \alpha_{jt} + \alpha_{st} + \sum_{\substack{b \in B \\ b \neq (9-11C)}} \beta_b D^b_{z(s)t} + \lambda X_{z(s)t} + \varepsilon_{ijz(s)t}, \tag{1}$$

where i indexes manufacturing plants, j indexes industries, $z(s)$ indexes the ZIP code z in state s where the plant is located, and t indexes years. Our plant-year panel spans the time period of 1977-2018. The main independent variables, D^b, capture the number of days in a given ZIP code and year whose average daily temperature is within a certain bin b. Our panel specification follows the approach of Deschênes and Greenstone (2011), which has been commonly employed in estimating temperature impacts as it allows arbitrary non-linear relationships between temperature and outcome variables.[14] We divide the temperature distribution into 11 bins of 3°C each, ranging from -6°C to 26°C, plus two additional bins capturing average daily temperatures below -6°C and above 26°C,

[13] See, e.g., Auffhammer (2018), Burke and Emerick (2016), and Blanc and Schlenker (2017) for a comprehensive discussion of each method, as well as their advantages and challenges.

[14] See Zhang et al. (2018), Heutel et al. (2021) for applications of the same methodology.

Electronic copy available at: https://ssrn.com/abstract=4526111

respectively. We exclude the median temperature bin 9°C-11°C in all specifications. The estimated β_b coefficients should be interpreted as the effect of an additional day with average temperature in a certain bin relative to an additional day with average temperature of 9°C-11°C. To account for geographical correlation in the error term, we cluster standard errors at state-level in all specifications.[15] In addition, when examining outcome variables obtained from ASM/CMF (i.e., energy costs, productivity, and employment), regressions are estimated using ASM sample weights.

Because plants have a fixed location over time, the inclusion of plant fixed effects (α_i) implies that the impact of temperature on outcomes is identified by deviations from plant-location specific means. As such, we think of these yearly temperature shocks as random "weather" draws from the "climate" distribution in a given geographical area, and therefore as plausibly exogenous to the outcomes of interest (Dell et al., 2014). We layer additional fixed effects step-by-step to absorb potential time-varying industry and geographic dynamics that might confound our key estimates. We first include 4-digit NAICS industry fixed effects interacted with year fixed effects to absorb any aggregate trends at industry-level experienced by US manufacturing plants. We then add in equation (1) state fixed effects interacted with year fixed effects to capture common trends in different areas of the US, which helps to ensure that the response of manufacturing to temperature shocks is identified by idiosyncratic local shocks.

Notice that temperature shocks can affect local manufacturing activity in two ways. First, they can affect the input costs and the production processes of plants, for example by increasing energy consumption, increasing maintenance costs of machinery and equipment or affecting the productivity of workers. We think of this set of forces as manufacturing *supply* shocks. Additionally, temperature shocks can affect local demand from consumers, for example via their impact on the profitability of local agriculture (Burke and Emerick, 2016). In the context of US manufacturing studied in this paper, we think of each US ZIP code or county as a small open economy and manufacturing as a tradable sector whose demand is geographically sparse across the US and the rest of the world, and thus relatively independent from local demand shocks. Under this assumption, supply forces are likely to be the major driver of the impact of temperature on manufacturing outcomes. We test this assumption in the data by studying how temperature shocks affect energy costs and labor productivity, which are both observable in our data.

Temperature shocks might be associated with precipitation or extreme weather events, and thus affect manufacturing outcomes via this association. We explore this relationship in Figure IV, which reports the effect of an additional day with average temperature within each bin on average precipitation and the incidence of extreme weather events recorded in

[15] Clustering at the state level is more conservative relative to clustering at finer geographic units, such as at the county level, since it allows standard errors to correlate within much larger geographic areas. All our results are robust (and more significant) when we cluster at the county level.

Electronic copy available at: https://ssrn.com/abstract=4526111

the aforementioned SHELDUS database. Perhaps unsurprisingly, additional hot days are associated with lower average precipitation and lower probability of floods. Additional hot days are mechanically associated with a higher probability of droughts and heatwaves, which are themselves defined based on prolonged occurrence of high temperature days. The effect of temperatures on tornados and hurricanes are small and mostly insignificant. We augment equation (1) with a set of time-varying controls $X_{z(s)t}$ which include average precipitation and the occurrence of extreme weather events that are not mechanically associated with temperature, mainly hurricanes and tornados.

III.B Long differences approach to study the effects of climate change

To study the long-run response of manufacturing activity to changes in average temperatures, we aggregate data at the county level and estimate the following long difference specification:

$$\begin{aligned} \Delta y_{c(s),2010s-1980s} &= \alpha_s + \beta_1 \Delta CDD_{c(s),2010s-1980s} + \beta_2 \Delta HDD_{c(s),2010s-1980s} \\ &+ \lambda X_{c(s)} + u_{c(s)} \end{aligned} \tag{2}$$

To estimate equation (2), we construct decadal averages of yearly data for both the manufacturing outcome variables and the temperature variables in the 1980-1989 decade and in the 2010-2019 decade in each county c. Long run differences are then calculated by subtracting the decadal average of 1980-1989 from the decadal average of 2010-2019.

Our choice of start- and end-point is motivated by three observations. First, as outlined in section II.B, the significant emergence of changes in temperature relative to historical averages occurred after 1981. Second, prior literature studying economic adaptation to long-run changes in temperature has also focused on the post-1980 period, noting that warming trends in the US after the 1980s have been larger than those observed in earlier periods (Burke and Emerick, 2016). Third, as explained in section II.A, the US Census LBD data provides consistent coverage of manufacturing activity for the period 1980-2019. Thus, we think of the forty year period 1980-2019 as long enough to capture significant changes in the average climate of each location.

In equation (2), we use two parsimonious measures of temperature: cooling degree days (CDD) and heating degree days (HDD). These are standard measures meant to capture the energy required to keep temperature at a baseline level, and have the advantage of capturing the non-linear impact of extreme temperature variation. Daily CDD is defined as the difference in degrees between the average daily temperature in a location and 18°C – the baseline temperature at which no heating or cooling is necessary – conditional on

Electronic copy available at: https://ssrn.com/abstract=4526111

the average daily temperature being above 18°C.[16] For each county, we compute CDD as the sum of all $CDDs$ over a year. $HDDs$ are defined in the same way for days with average daily temperature below 18°C.

Equation (2) includes state fixed effects, which implies that the relevant variation identifying the coefficients β_1 and β_2 originate from within-state differences in climate trends across counties. The inclusion of state fixed effects removes any role of unobservable state-level trends. However, a potential concern is whether their inclusion also removes most of the relevant variation in long-term changes in climate. We investigate this concern in Figure V and Figure VI.

Figure V reports the distribution of long run changes in decadal averages of HDD and CDD. Panel (a) reports the distribution of these two variables in the raw data. As shown, between the 1980s and the 2010s, most US counties experienced an increase in average yearly cooling degree days, or degree days above 18°C, while the changes in heating degree days were mostly negative in the same period. This is consistent with a significant warming trend across US counties during the last four decades. Panel (b) reports the distribution of long run changes in decadal averages of HDD and CDD in deviation from state averages. As shown, even net of state trends, there is still significant variation in degree days across counties. For example, a standard deviation in the raw distribution of long run changes in CDD corresponds to 120 degree days (see Table I), while after removing state fixed effects a standard deviation in the same variable corresponds to 57 degree days. We rely on this variation in our estimates of long-run effects of changes in average climate on manufacturing activity. Figure VI reports the geographical distribution of these long-run changes in degree days in deviations from state-specific averages.

The key identifying assumption in equation (2) is that differential changes in degree days observed over the last four decades in each county are uncorrelated with other local trends that might also affect the outcomes of interest. State fixed effects reduce the role of unobservables by removing state-level aggregate trends. Still, a potential concern is that long run changes in temperature might be correlated with unobservable county-level trends. In support of empirical approaches similar to the one described in equation (2), previous literature in environmental economics has underlined how "recent evidence from the physical sciences suggests that the large differential warming trends observed over the United States over the past few decades are likely due to natural climate variability" rather than trends in local emissions or changes in local land use (Burke and Emerick, 2016). In support of this assumption, in Panel A of Table A.2 we show that long-run changes in

[16] This implies that a day with average temperature of 20°C will correspond to 2 CDD and a day with average temperature of 12°C to 0 CDD. See, for instance, Heutel et al. (2021) or Zivin and Kahn (2016) for applications of $CDDs$ constructed relative to a baseline temperature of 65°F and Burke and Emerick (2016) for a CDD-type measure adjusted to the importance of temperature deviations during growing seasons in agriculture. See also the discussion by the National Oceanic and Atmospheric Service, `https://www.weather.gov/key/climate_heat_cool`.

Electronic copy available at: https://ssrn.com/abstract=4526111

average temperatures are not strongly correlated with county-level initial characteristics, including population, per capita income, and share of college graduates among the adult population. We also check the correlation of long-run increases in temperature with exposure to shocks that might be particularly important for US manufacturing in the period under study, such as import competition from China (Autor et al., 2013), finding no significant correlation (column (4)).

In Panel B of Table A.2 we test the correlation of long-run changes in temperature with long-run changes in frequency of reported natural disasters such as floods, droughts, heatwaves hurricanes, and tornadoes , as well as long-run changes in average precipitation. Overall, we find non-significant correlations in the expected direction between changes in temperatures and changes in the frequency of natural disasters. The correlation with average precipitation is instead negative and strongly significant. In the empirical analysis we include in equation (2) the initial county characteristics reported in Panel A and also control for long-run changes in the natural hazards that are not mechanically a function of temperature (hurricanes and tornadoes) and average precipitation. We show that the magnitude of the point estimates is stable after the inclusion of these controls.

IV Empirical Results

IV.A Short-run response to temperature shocks

In this section we study the short-run effects of temperature shocks on manufacturing outcomes. We start by focusing on two outcomes plausibly affected by an increase in hot days relative to the climate usually experienced in a given location: energy costs and productivity of manufacturing plants. Next, we study the impact of temperature shocks on both the intensive margin (plant size) and the extensive margin (entry and exit) of manufacturing activity.

Energy costs. Higher than usual temperatures can increase energy costs of manufacturing production in several ways. First, due to an increase in the demand for electricity necessary to cool down work environments via air conditioning, as well as to cool down machinery and equipment used in production. Second, higher temperatures can negatively affect the efficiency of energy production systems and transmission: an increase in hot days implies that power plants need to be cooled down more often or cannot operate due to decreases in water availability, and electrons move slower inside transmission lines at higher temperatures (Bartos et al., 2016).[17]

We estimate equation (1) when the outcome variable is energy costs at the plant level.

[17] Although the effect of temperature on energy production systems is relevant in the aggregate, it does not necessarily manifest in the same ZIP code of the manufacturing plant using the energy (because power plants might be located elsewhere), and thus it is less likely to be captured by our empirical strategy.

Electronic copy available at: https://ssrn.com/abstract=4526111

The results are reported in Table II and visualized in Figure VII. Energy costs are defined as the monetary value of expenses in electricity and fuel divided by the value of shipments at plant level. All coefficients are multiplied by 100 to facilitate readability, so the point estimates should be interpreted as the effect of 100 additional days in a given temperature bin relative to the omitted benchmark bins experienced by a given plant-location. We find that plants experiencing additional days with average temperature above 18°C experience statistically significant increases in energy costs. The effect is monotonically increasing in temperature bins and the magnitude of the coefficients implies that a year with 100 additional days in the temperature bins above 18°C would generate a 0.2 percentage points increase in energy costs as a share of value of shipment. This corresponds to about 9% of the sample average in the outcome variable. Alternatively, a one-standard-deviation increase in the number of days above 26°C (31 days) would generate a 0.07 percentage points increase in energy costs as a share of value of shipment, or 2.8% of the sample average.

We find non significant effects of additional cold days on energy costs, possibly with the exception of the coldest temperature bin. The asymmetry between the effects of additional hot days vs cold days on energy costs is prima facie surprising. One element that contributes to explain this finding is that – despite being less seasonal than residential or commercial energy consumption – industrial energy consumption displays large differences between summer and winter months. Data from the US Energy Information Administration (EIA) shows that during the summer months, average consumption in kilowatthours has been 9.6% higher than during the winter months in the years since 2002. The summer period is also a period of higher energy prices faced by the industrial sector in the US. EIA data shows that average electricity prices in cents per kilowatthour for the industrial sector have been on average 10.5% higher during the summer months relative to the winter months between 2002 and 2023.[18] Taken together, these facts indicate that additional hot days during the summer period generate higher marginal increases in energy costs for US industrial firms than additional cold days in the winter period.

Next, we investigate the effect of temperature shocks on energy costs of small vs large plants in Figure VIII. Throughout the paper, we define small vs large plants based on number of employees. Results are similar depending on whether we consider as small plants those with less than 20 or 50 employees. Panels (a) and (b) of Figure VIII document that the effects of temperature shocks on energy costs are concentrated among small plants. Large manufacturing plants – and especially those above 50 employees – seem to be largely immune from the effects of temperature shocks on energy costs.

There are several potential mechanisms that can explain this result. First, small firms

[18] Calculations done by the authors based on Electric Power Monthly dataset available at `https://www.eia.gov/electricity/data.php`. We use June to August as summer months, January to March as winter months.

Electronic copy available at: https://ssrn.com/abstract=4526111

tend to face higher energy prices than large firms. The latest Manufacturing Energy Consumption Survey (MECS) run in 2018 on a nationally representative sample of US manufacturing establishments shows that establishments with under 50 employees face electricity prices (in USD per Million Btu) that are 46% higher than those faced by manufacturing establishments with 250 employees and above.[19] Large industrial establishments are able to negotiate better prices from electricity suppliers for several reasons, including: (i) they use more electricity and can receive it at higher voltages, making transmission less expensive; (ii) their demand is less seasonal and less volatile during the day, so they can receive discounts by lowering their use during consumption peaks by retail customers that put the electric grid under stress.

Another potential explanation for the heterogeneous effects of additional hot days on energy costs of establishments of different size is that large plants might be more likely to have implemented energy-saving technologies or operate with capital – machinery, equipment, buildings – of higher "quality" that is less sensible to temperature shocks. For example, larger plants might have better insulated buildings or newer machinery and equipment used in production that are more energy efficient and less prone to overheating, thus requiring less cooling of production spaces. We return to this point in section V.

Productivity. A large existing literature has documented a negative relationship between temperature and labor productivity (see, among others, Graff Zivin and Neidell (2014), Heal and Park (2013), Hsiang (2010), and Somanathan et al. (2021)). Rising temperatures can affect manufacturing productivity via their effect on both the performance of workers and the productivity of machinery and equipment. The effect of temperature on workers' productivity can arise due to fatigue and lower ability to focus, as well as absenteeism. Stricter safety standards have increased the amount of protective gear necessary in manufacturing workplaces over time, amplifying the exhaustion of performing the same task at a higher temperature. Another amplifying effect might arise from the faster physical pace or longer shifts set by manufacturing plants in order to meet production goals and remain competitive in a global market. On the other hand, direct evidence on the effects of temperature on the performance of machinery and equipment is more sparse, though Zhang et al. (2018) show suggestive evidence that higher temperatures lower capital productivity for Chinese manufacturers. In what follows we document similar negative effects of temperature shocks on productivity for US manufacturing plants, and provide new evidence on how these effects differ for small vs large plants.

The results of estimating equation (1) when the outcome variables are different measures of productivity are reported in Table III and visualized in Figure IX. We use two measures of productivity: total factor productivity (TFP) and labor productivity at the

[19] See Table 7.5, MECS publication, released on September 2021.

Electronic copy available at: https://ssrn.com/abstract=4526111

plant level, both in logs. TFP is computed as the plant-level Solow residual. Labor productivity is defined as valued added divided by total number of employee-hours worked. Point estimates should be interpreted as the effect of 100 additional days in a given temperature bin relative to the average climate experienced by a given plant-location. We find a negative and monotonic effect of temperature on both measures of productivity, with additional days in hotter bins leading to lower productivity. The positive effects on additional cold days are small and mostly not statistically significant, while plants experiencing additional hot days experience significant declines in productivity. The magnitude of the coefficients imply that a year with 100 additional days in the temperature bins above 18°C would generate a 4 percentage points decline in TFP and a 7 percentage point decline in labor productivity as measured by value added per hour worked. Alternatively, if we look at the extremely hot days, a one-standard-deviation increase in the number of days above 26°C (31 days) would generate a 0.9 percentage points decline in TFP and a 2.41 percentage point decline in labor productivity.[20]

Next, in Figure X, we report the results when splitting the sample between small vs large plants. Independently of the threshold used to define small plants, we find that higher than usual temperatures are associated with large and significant declines in the productivity of small plants. On the other hand, the effects of temperature shocks on the productivity of large manufacturing plants are small and mostly non statistically significant. Mechanisms that can rationalize these heterogeneous effects of temperature shocks on plant productivity include heterogeneity in the type of labor and capital used by plants of different size. Larger plants tend to produce with physical capital whose performance is less affected by abnormal temperatures. Examples include higher probability to have temperature control systems (Zivin and Kahn, 2016), or better insulated work environments. Differences in the type of labor force employed in large vs small plants might also play a role. For examples, large plants are matched with more productive, more motivated workers whose performance is less affected by temperature shocks.

Size, Entry and Exit. After documenting contemporaneous effects of temperature shocks on energy costs and productivity, we now focus on studying whether US manufacturing plants respond to temperature shocks via the intensive margin (e.g. by increasing or decreasing their size) or via the extensive margin (e.g. by deciding to enter or exit

[20]Because energy is an input in production, the increase in energy costs documented above could mechanically generate a decline in value added, and thus in TFP or labor productivity measured as value added per worker. We checked this potential explanation of the productivity results by estimating equation (1) using alternative measures of productivity that are not a function of energy costs, such as: value of shipments per worker or per hour worked, and value added per worker or per hour worked where value added is constructed without including energy among inputs. We find similar results using these alternative measures, which indicates that the effect of temperature shocks on productivity is not mechanically driven by the effect of temperature shocks on energy costs.

Electronic copy available at: https://ssrn.com/abstract=4526111

certain locations).

We start by studying the effect of temperature shocks on plant size, as measured by total number of employee-hours. The results are reported in Table IV and visualized in Figure XI. As shown, we find no contemporaneous response to additional hot days relative to what usually experienced by a given plant, and a positive but noisy response to additional cold days. Figure XII also documents that small and large plants are similarly non responsive to temperature shocks on the intensive margin.

Next, we focus on the extensive margin, and in particular on the decision of a given plant to enter or exit a given ZIP code. To this end, we use data from the LBD described in section II.A, which tracks all manufacturing establishments along with their location and size over time. When estimating equation (1), we define entry of plant i in ZIP code z during year t as a dummy equal to 1 if plant i has no employment in year $t-1$ and positive employment in year t. We define exit in year t as a dummy equal to 1 if plant i has positive employment in the LBD in year t but no recorded employment in year $t+1$. This is because plants that are in operation for a fraction of a year are still recorded in the LBD for that year, so our definition ensures that we are capturing the contemporaneous relationship between temperature shocks and exit decisions.

The results for entry are reported in columns (1) to (4) of Table V and visualized in Figure XIII. We find that entry is more likely to occur in years with additional "median" temperature days, while the probability of entry declines with both additional cold and hot days. As shown, higher than usual number of hot days are detrimental for the opening of a new plant in a given location, though the economic magnitude of the coefficient on the contemporaneous effects is relatively small. The magnitude of the estimated coefficients imply that a year with 100 additional days in the temperature bins above 18°C corresponds to a 1 percentage point decline in the probability of plant opening in that year. This is an economically large effect when considering the average entry rate in our sample is 0.07 (Table I).

Figure XIV shows that the effects of temperature shocks on entry are concentrated on plants with less than 20 or 50 employees. The effects on large plants follow a pattern similar to small plants across temperature bins, but are smaller in magnitude and noisier – in part because most manufacturing plants are small at the time of opening.

The results for exit are reported in columns (5) to (8) of Table V and visualized in Figure XV. We find mostly small in magnitude and non significant effects of temperature shocks on exit. The probability of exit monotonically increases with temperature bins above 18°C but even estimates on highest temperature realizations are not statistically significant. Notice that the majority of exit events in the LBD occur after plants decrease in size and thus enter into the category of small plants. This implies that when studying the heterogeneous effects of temperature shocks on exit by plant size we can only estimate the saturated model described in equation (1) for small plants. These results are reported

Electronic copy available at: https://ssrn.com/abstract=4526111

in Figure XVI, and show a similar pattern as Figure XV.

Discussion of short-run response to temperature shocks. There are two key findings that emerge from our estimates of the short run effects of temperature shocks on manufacturing outcomes. First, input costs associated with temperature management (such as energy costs) and productivity react to contemporaneous temperature shocks. In particular, higher than usual temperatures increase energy spending and decrease plant productivity. However, in the short run, plants do not seem to respond to temperature shocks via contemporaneously adjusting employment in a significant way. We also do not find evidence that these temperature shocks significantly affect the probability of exit, while hotter than usual temperatures seem to dissuade entry. Second, the effects described above are driven by small manufacturing plants, while we document mostly small and non significant effects for large establishments.

IV.B Long-run response to changes in temperature

The short run response to temperature shocks documented in section IV.A indicates that small plants incur significant additional energy costs and lower productivity in hotter than usual years. However, these effects do not trigger significant contemporaneous adjustments on the intensive or extensive margin, possibly with the exception of a lower likelihood of entering. It is plausible that key industrial decisions such as scaling back on the size of an existing plant or exiting a given market are not driven by yearly weather shocks, especially if such shocks are interpreted as idiosyncratic and therefore likely to revert in following years. On the other hand, the cumulative effect of several years of hotter than usual weather might push managers to respond on these margins. This is because a series of deviations from past temperatures might indicate a shift in the "climate" distribution from which "weather" events are drawn in a given geographical area.

To investigate the response to long-run changes in average temperatures in a given county we estimate equation (2) described in section III.B. This equation relates long-run changes in manufacturing activity with long-run changes in average temperatures between the 1980s decade and the 2010s decade. As discussed in section II.B, the US has experienced a large increase in average temperatures between the 1980s and 2010s, with substantial variation even across counties within the same state. Because the short-run effects indicate that there are significant heterogeneous effects of temperature shocks across plants of different size, in what follows we investigate the effects of long run changes in average temperatures separately for small vs large plants.

Number of plants and employment. We start by studying the effect of long-run changes in temperature on long-run changes in the number of plants of different size in a given county. The results are visualized in the top panel of Figure XVII and reported in

Electronic copy available at: https://ssrn.com/abstract=4526111

columns (1) to (4) of Table VI, panels A and B. The point estimate reported in column (2) of Panel A implies that counties that in the 2010s decade had a standard deviation higher increase in temperature – about 100 degree days per year above 18°C – relative to the 1980s decade experienced a 4.2 percent larger decline in the number of manufacturing plants with less than 20 employees. This effect remains stable in magnitude and statistically significant when controlling for the set of initial county characteristics described in section III.B, which lends support to the empirical strategy.

The relative decline in the number of small plants does not translate into an increase in the number of large plants, as shown in columns (3) and (4). This indicates that the documented effect on small plants is not driven by a higher transition from small to large plants (i.e. small plants becoming large plants) in areas that are getting warmer at a faster pace. We also document that changes in cooling degree days between the 1980s and the 2010s had smaller and non statistically significant effects on the number of manufacturing plants in a given county. In Panel B, we replicate the analysis using 50 employees as a threshold to define large plants, finding similar results.

Next, we investigate the long-run effects of higher average temperatures on number of workers employed by small vs large plants. The results are visualized in the bottom panel of Figure XVII and reported in columns (5) to (8) of Table VI. We find negative and significant effects of long-run changes in average temperatures on employment in small plants. The estimated coefficient in column (6) of Panel A indicates that counties that in the 2010s decade had a standard deviation higher increase in temperature – about 100 degree days per year above 18°C – relative to the 1980s decade experienced a 5.9 percent larger decline in the number of workers employed by manufacturing plants with less than 20 employees. The effects on number of workers in large plants are positive and marginally significant, indicating that while the number of large plants does not increase, these plants are able to absorb the labor force lost by smaller manufacturing establishments.

Finally, in Table VII we document the overall effect of long-run increases in temperatures on total number of plants, overall employment and average plant size in a given county. We find negative but noisy estimates on the effects on total number of plants, and positive but non significant effects on total employment. Columns (5) and (6) show instead positive and significant effects on average plant size. Overall, these results indicate that counties with faster warming temperatures did not experience significant major changes in total employment, but mostly a reallocation of production from small to large plants, leading to a 7.7 percent higher increase in average plant size for a standard deviation higher increase in average temperatures over the last four decades.

Concentration of manufacturing activity. The results presented in Tables VI and VII indicate that, over the long run, areas getting warmer at a faster pace experienced larger declines in the number of small plants and no effect on the number of large plants.

Electronic copy available at: https://ssrn.com/abstract=4526111

In this section we study the impact of long-run changes in temperature on local concentration in manufacturing activity. In particular, we focus on the percent of employment concentrated in the top-5 largest plants in a given county, and the Herfindahl-Hirschman Index (HHI). We compute the Herfindahl-Hirschman Index as the sum of squared values of the employment shares of each plant in a given county. The index thus captures the amount of concentration in the employment share across plants, with higher values indicating higher concentration.

The results are reported in Table VIII. The point estimates indicate positive and significant effects of long run changes in average temperatures on industrial concentration. In particular, we find that counties that in the 2010s decade had a standard deviation higher increase in temperature – about 100 degree days per year above 18°C – relative to the 1980s decade experienced a 1.4 percentage points larger increase in the share of employment concentrated in the top 5 largest plants, and a 5.1 percent larger increase in the Herfindahl-Hirschman concentration index. Overall, the results indicate that faster warming in the last four decades has led to higher concentration of industrial activity among larger plants. We discuss potential mechanisms behind this result in the next section.

V Discussion of mechanisms

The finding that faster warming has led to higher concentration of manufacturing activity among larger plants suggests that such plants might be better equipped for long-run adaptation to climate change. In this preliminary version of this section we list potential economic mechanisms that can rationalize the main results of the paper. We also explain how we are testing these mechanisms empirically.

i. Hedging. Large firms might be naturally better hedged to absorb weather shocks, even when they occur at higher frequency due to climate change. For example, Castro-Vincenzi (2022) documents how car companies are able to partly absorb weather shocks – like floods – by reallocating production from affected plants to non-affected plants. This hedging strategy requires firms to keep spare capacity in each location, which multi-plant large firms are more likely to be able to afford.

In order to test this mechanism in the data, we plan to investigate whether the relative long-run decline in number of small establishments differs depending on whether these small establishments are part of a multi-plant firm or not. More specifically, we estimate equation (2) using two outcome variables: the change in the number of small single-plant establishments, and the change in the number of small establishments that are part of a multi-plant firm. Preliminary results show that, while single- and multi-unit small plants are similarly affected by temperature shocks in terms of their energy costs and

Electronic copy available at: https://ssrn.com/abstract=4526111

productivity, the negative impact of long-run changes in temperatures on employment is larger for single-unit plants relative to plants that are part of a multi-unit firm.

ii. Access to finance. Another potential mechanism linking firm size with better adaptation to climate change is that large firms might also have better access to external finance. This would allow them to use available credit lines to cope with weather shocks, reducing the need to downscale employment or close plants. For example, using data from Brazil, Albert et al. (2021) document how access to finance helped drought affected municipalities to insure themselves against the negative impact of weather shocks via capital inflows from regions connected via the bank branch network. Easier access to external finance also facilitates investments in long term projects necessary to make their production process less sensitive to climate change. In the context of agriculture, Rajan and Ramcharan (2023) document that access to bank finance facilitated the long-run adjustment to the 1949-1957 drought in the US. They show that counties with initially better access to external finance experienced lower out-migration, and their agricultural sector was better able to adapt via investments in irrigation, drought-tolerant crops and mechanization.

To test this mechanism, we plan to use data from the Quarterly Financial Reports collected by the US Census, which allows us to observe access to finance at the firm level in each Census year. We plan to match this data with the LBD dataset and then: (i) document if significant differences exist in access to external finance between small and large firms in our sample of US manufacturers and during the period under study; and (ii) whether the long run effects on both number of establishments and employment in large firms depend on their ability to access external finance.

iii. Managerial skills. Large firms might also have better trained managers who understand the change in firm exposure to climate risk and invest in adaptation. This hypothesis relies on two findings documented in the existing literature. First, there is evidence that large firms tend to be better managed. For example, Bloom et al. (2019) document large dispersion in management practices across US manufacturing plants, and show that the diffusion of "structured" management practices is strongly correlated with both plant and firm size as captured by number of employees (Figure A2 in Bloom et al. 2019).

Second, the literature has also established that better managed firms are less energy intensive and more productive (Bloom et al. 2010, Martin et al. 2012), and that more attentive managers are able to offset some of the adverse effects of warmer temperatures on productivity by means of task reallocation (Adhvaryu et al. 2022). Within our setting, examples of investments into adaptation include the adoption of technologies that reduce the effect of temperature on labor productivity, such as automated warehouse management systems, the update of buildings and machinery so that they can better withstand higher temperatures or natural disasters, and the adoption of general energy-saving tech-

Electronic copy available at: https://ssrn.com/abstract=4526111

nologies such as computer systems to control major energy-using equipment. Data from the Manufacturing Energy Consumption Survey (MECS) shows that the share of manufacturing plants engaging in energy management activities in the US has increased from 40% in 2002 to 48% in 2018, the year of the last survey.

VI Concluding remarks

In this paper, we study the short-run and long-run implications of temperature variation on U.S. manufacturing establishments. We find extreme temperatures, and particularly warmer temperatures, to result in short-term increases in the cost of inputs associated with temperature management (electricity and fuel costs), short-term declines in productivity, and short-term deterrence of entry. While these effects are concentrated in small manufacturing plants, large establishments are mostly unaffected. In the long run, we find that areas faced with greater increases in average temperatures between the 1980s and the 2010s experienced larger declines in the number of small plants. While the number of large plants did not increase, large plants were able to absorb at least part of the labor force lost by smaller manufacturing establishments.

Taken together, the results are consistent with large firms being better equipped for long-run adaptation to climate change, which leads to higher industry concentration. Among the various channels that may help large plants adapt better are their ability to hedge in order to absorb weather shocks, better access to external finance, and higher managerial quality. This highlights that recent increases in industry concentration are note solely due to technological and political factors, but that climate change plays an important role as well.

The results also raise the question of whether higher concentration of employment within large firms is "good or bad" for the local economies more affected by a warming climate. Our paper does not estimate the welfare implications of manufacturing concentration. However, some of our results speak to this debate. For example, the presence of large firms with the means to adapt to long-run changes in climate could be an important factor in preserving employment locally and limiting out-migration. Indeed, our results show that faster warming leads to a reallocation of employment from small to large firms but no significant changes in the overall employment at the county level. On the other hand, the differential effect of temperature on small firms might have detrimental effects on outcomes associated with small scale firms – such as "radical" innovations, as some of the literature has suggested (Prusa and Schmitz Jr, 1991) – or even become a barrier to entrepreneurship in certain regions. We think of these issues as important potential avenues for future research.

Electronic copy available at: https://ssrn.com/abstract=4526111

References

Acharya, V. V., T. Johnson, S. Sundaresan, and T. Tomunen (2022). Is physical climate risk priced? evidence from regional variation in exposure to heat stress. Technical report, National Bureau of Economic Research.

Addoum, J. M., D. T. Ng, and A. Ortiz-Bobea (2020). Temperature shocks and establishment sales. *The Review of Financial Studies 33*(3), 1331–1366.

Addoum, J. M., D. T. Ng, and A. Ortiz-Bobea (2021). Temperature shocks and industry earnings news. *Available at SSRN 3480695*.

Adhvaryu, A., N. Kala, and A. Nyshadham (2022). Management and shocks to worker productivity. *Journal of Political Economy 130*(1), 1–47.

Albert, C., P. Bustos, and J. Ponticelli (2021). The effects of climate change on labor and capital reallocation. Technical report, National Bureau of Economic Research.

Alexander, L. and J. Eberly (2018). Investment hollowing out. *IMF Economic Review 66*, 5–30.

Auffhammer, M. (2018). Quantifying economic damages from climate change. *Journal of Economic Perspectives 32*(4), 33–52.

Auffhammer, M., S. M. Hsiang, W. Schlenker, and A. Sobel (2013). Using weather data and climate model output in economic analyses of climate change. *Review of Environmental Economics and Policy*.

Auffhammer, M., V. Ramanathan, and J. R. Vincent (2006). Integrated model shows that atmospheric brown clouds and greenhouse gases have reduced rice harvests in india. *Proceedings of the National Academy of Sciences 103*(52), 19668–19672.

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. V. Reenen (2017). Concentrating on the fall of the labor share. *American Economic Review 107*(5), 180–185.

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020). The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics 135*(2), 645–709.

Autor, D. H., D. Dorn, and G. H. Hanson (2013). The china syndrome: Local labor market effects of import competition in the united states. *American economic review 103*(6), 2121–2168.

Bartos, M., M. Chester, N. Johnson, B. Gorman, D. Eisenberg, I. Linkov, and M. Bates (2016). Impacts of rising air temperatures on electric transmission ampacity and peak electricity load in the united states. *Environmental Research Letters 11*(11), 114008.

Blanc, E. and W. Schlenker (2017). The use of panel models in assessments of climate impacts on agriculture. *Review of Environmental Economics and Policy*.

Bloom, N., E. Brynjolfsson, L. Foster, R. Jarmin, M. Patnaik, I. Saporta-Eksten, and J. Van Reenen (2019). What drives differences in management practices? *American Economic Review 109*(5), 1648–1683.

Electronic copy available at: https://ssrn.com/abstract=4526111

Bloom, N., C. Genakos, R. Martin, and R. Sadun (2010). Modern management: Good for the environment or just hot air? *The economic journal 120*(544), 551–572.

Burke, M. and K. Emerick (2016). Adaptation to climate change: Evidence from us agriculture. *American Economic Journal: Economic Policy 8*(3), 106–40.

Burke, M., S. M. Hsiang, and E. Miguel (2015). Global non-linear effect of temperature on economic production. *Nature 527*(7577), 235–239.

Cahill, N., S. Rahmstorf, and A. C. Parnell (2015). Change points of global temperature. *Environmental Research Letters 10*(8), 084002.

Castro-Vincenzi, J. M. (2022). Climate hazards and resilience in the global car industry. *Working paper*.

Chen, X. and L. Yang (2019). Temperature and industrial output: Firm-level evidence from china. *Journal of Environmental Economics and Management 95*, 257–274.

Colacito, R., B. Hoffmann, and T. Phan (2019). Temperature and growth: A panel analysis of the united states. *Journal of Money, Credit and Banking 51*(2-3), 313–368.

Crouzet, N. and J. C. Eberly (2021). Rents and intangible capital: A q+ framework. Technical report, National Bureau of Economic Research.

Custodio, C., M. A. Ferreira, E. Garcia-Appendini, and A. Lam (2022). How does climate change affect firm sales? identifying supply effects. *Identifying Supply Effects (June 30, 2022)*.

Davis, D. R. and D. E. Weinstein (2002). Bones, bombs, and break points: the geography of economic activity. *American economic review 92*(5), 1269–1289.

De Loecker, J., J. Eeckhout, and G. Unger (2020). The rise of market power and the macroeconomic implications. *The Quarterly Journal of Economics 135*(2), 561–644.

Dell, M., B. F. Jones, and B. A. Olken (2009). Temperature and income: reconciling new cross-sectional and panel estimates. *American Economic Review 99*(2), 198–204.

Dell, M., B. F. Jones, and B. A. Olken (2012). Temperature shocks and economic growth: Evidence from the last half century. *American Economic Journal: Macroeconomics 4*(3), 66–95.

Dell, M., B. F. Jones, and B. A. Olken (2014). What do we learn from the weather? the new climate-economy literature. *Journal of Economic Literature 52*(3), 740–98.

Deryugina, T. and S. M. Hsiang (2014). Does the environment still matter? daily temperature and income in the united states. Technical report, National Bureau of Economic Research.

Deschênes, O. and M. Greenstone (2011). Climate change, mortality, and adaptation: Evidence from annual fluctuations in weather in the us. *American Economic Journal: Applied Economics 3*(4), 152–185.

Desmet, K. and E. Rossi-Hansberg (2021). The economic impact of climate change over time and space. *NBER Reporter* (4), 16–20.

Electronic copy available at: https://ssrn.com/abstract=4526111

Ersahin, N., R. M. Irani, and K. Waldock (2021). Can strong creditors inhibit entrepreneurial activity? *The Review of Financial Studies 34*(4), 1661–1698.

Feenstra, R. C. and D. E. Weinstein (2017). Globalization, markups, and us welfare. *Journal of Political Economy 125*(4), 1040–1074.

Fisher, A. C., W. M. Hanemann, M. J. Roberts, and W. Schlenker (2012). The economic impacts of climate change: evidence from agricultural output and random fluctuations in weather: comment. *American Economic Review 102*(7), 3749–3760.

Foster, L., C. Grim, and J. Haltiwanger (2016). Reallocation in the great recession: cleansing or not? *Journal of Labor Economics 34*(S1), S293–S331.

Gallup, J. L., J. D. Sachs, and A. D. Mellinger (1999). Geography and economic development. *International regional science review 22*(2), 179–232.

Graff Zivin, J. and M. Neidell (2014). Temperature and the allocation of time: Implications for climate change. *Journal of Labor Economics 32*(1), 1–26.

Grullon, G., Y. Larkin, and R. Michaely (2019). Are us industries becoming more concentrated? *Review of Finance 23*(4), 697–743.

Gupta, R., E. Somanathan, and S. Dey (2017). Global warming and local air pollution have reduced wheat yields in india. *Climatic Change 140*(3-4), 593–604.

Gutiérrez, G. and T. Philippon (2017). Declining competition and investment in the us. Technical report, National Bureau of Economic Research.

Heal, G. and J. Park (2013). Feeling the heat: Temperature, physiology & the wealth of nations. Technical report, National Bureau of Economic Research.

Heutel, G., N. H. Miller, and D. Molitor (2021). Adaptation and the mortality effects of temperature across us climate regions. *The review of economics and statistics 103*(4), 740–753.

Hornbeck, R. (2012). The enduring impact of the american dust bowl: Short-and long-run adjustments to environmental catastrophe. *American Economic Review 102*(4), 1477–1507.

Hsiang, S., R. Kopp, A. Jina, J. Rising, M. Delgado, S. Mohan, D. Rasmussen, R. Muir-Wood, P. Wilson, M. Oppenheimer, et al. (2017). Estimating economic damage from climate change in the united states. *Science 356*(6345), 1362–1369.

Hsiang, S. and R. E. Kopp (2018). An economist's guide to climate change science. *Journal of Economic Perspectives 32*(4), 3–32.

Hsiang, S. M. (2010). Temperatures and cyclones strongly associated with economic production in the caribbean and central america. *Proceedings of the National Academy of sciences 107*(35), 15367–15372.

IPCC (2021). "Climate change 2021: The Physical Science Basis". *Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change 2*.

Electronic copy available at: https://ssrn.com/abstract=4526111

Jia, R., X. Ma, and V. W. Xie (2022). Expecting floods: Firm entry, employment, and aggregate implications. Technical report, National Bureau of Economic Research.

Jones, B. F. and B. A. Olken (2010). Climate shocks and exports. *American Economic Review 100*(2), 454–459.

Lin, Y., T. K. McDermott, and G. Michaels (2021). Cities and the sea level.

Lobell, D. B., W. Schlenker, and J. Costa-Roberts (2011). Climate trends and global crop production since 1980. *Science 333*(6042), 616–620.

Martin, R., M. Muûls, L. B. De Preux, and U. J. Wagner (2012). Anatomy of a paradox: Management practices, organizational structure and energy efficiency. *Journal of Environmental Economics and Management 63*(2), 208–223.

Miguel, E. and G. Roland (2006). The long run impact of bombing vietnam.

Olmstead, A. L. and P. W. Rhode (2011). Adapting north american wheat production to climatic challenges, 1839–2009. *Proceedings of the National Academy of sciences 108*(2), 480–485.

Ortiz-Bobea, A., E. Knippenberg, and R. G. Chambers (2018). Growing climatic sensitivity of us agriculture linked to technological change and regional specialization. *Science advances 4*(12), eaat4343.

Pankratz, N., R. Bauer, and J. Derwall (2023). Climate change, firm performance, and investor surprises. *Management Science*.

Pankratz, N. and C. Schiller (2021). Climate change and adaptation in global supply-chain networks. In *Proceedings of Paris December 2019 Finance Meeting EUROFIDAI-ESSEC, European Corporate Governance Institute–Finance Working Paper*, Number 775.

Prusa, T. J. and J. A. Schmitz Jr (1991). Are new firms an important source of innovation?: Evidence from the pc software industry. *Economics Letters 35*(3), 339–342.

Rajan, R. and R. Ramcharan (2023). Finance and climate resilience: Evidence from the long 1950s us drought. Technical report, National Bureau of Economic Research.

Samuelson, P. A. (1947). Welfare economics, foundations of economic analysis.

Schlenker, W. and M. J. Roberts (2009). Nonlinear temperature effects indicate severe damages to us crop yields under climate change. *Proceedings of the National Academy of sciences 106*(37), 15594–15598.

Somanathan, E., R. Somanathan, A. Sudarshan, and M. Tewari (2021). The impact of temperature on productivity and labor supply: Evidence from indian manufacturing. *Journal of Political Economy 129*(6), 1797–1827.

Viner, J. (1958). *The long view and the short: Studies in economic theory and policy*. Glencoe, Ill., Free P.

Electronic copy available at: https://ssrn.com/abstract=4526111

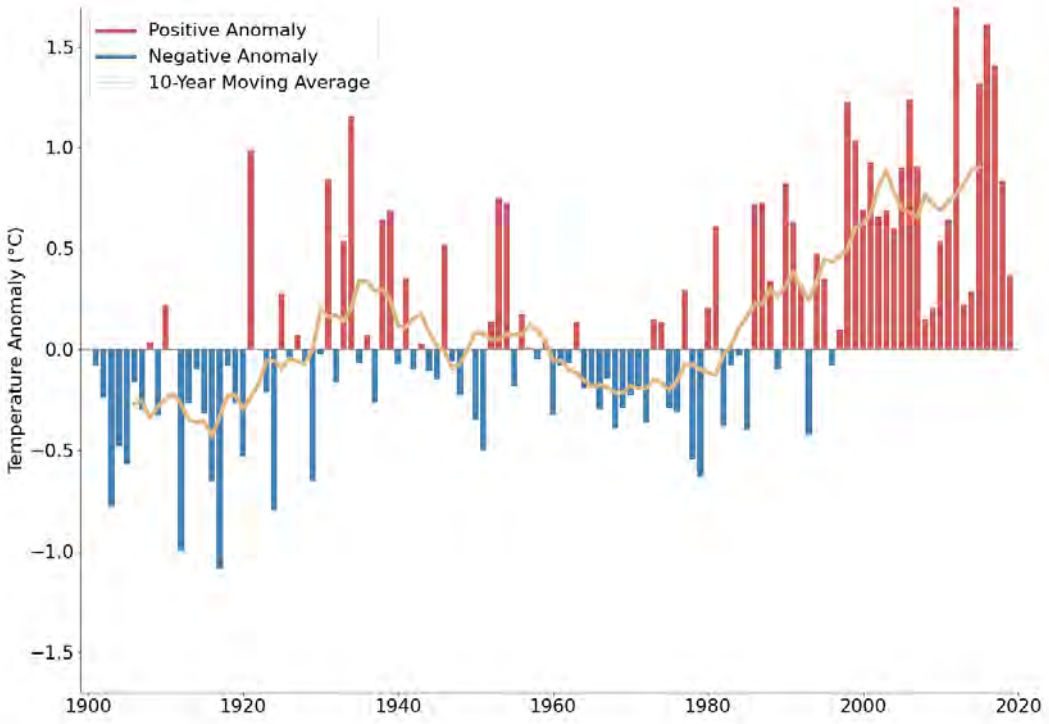
Zhang, P., O. Deschenes, K. Meng, and J. Zhang (2018). Temperature effects on productivity and factor reallocation: Evidence from a half million chinese manufacturing plants. *Journal of Environmental Economics and Management 88*, 1–17.

Zivin, J. G. and M. E. Kahn (2016). Industrial productivity in a hotter world: the aggregate implications of heterogeneous firm investment in air conditioning. Technical report, National Bureau of Economic Research.

Electronic copy available at: https://ssrn.com/abstract=4526111

VII Figures

Figure I: Temperature trend in the U.S.

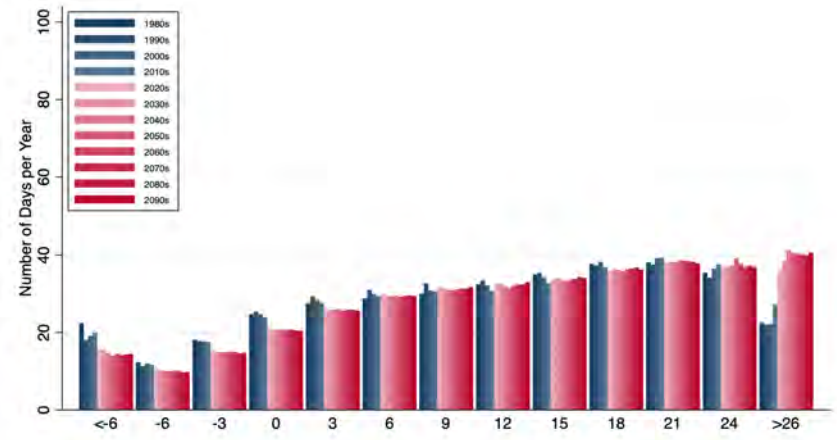


The figure shows annual and decadal temperature dynamics over the 1901-2019 period on the basis of temperature data obtained from the National Oceanic and Atmospheric Administration (NOAA). The underlying data covers 48 contiguous US states. Annual anomalies (red and blue bars) are defined as the difference between the annual average temperature across the 48 states and the average annual temperature over the 1901-2000 period. The moving average of the anomaly (line) is based on 10 years of anomaly observations centered around years [-4;5].

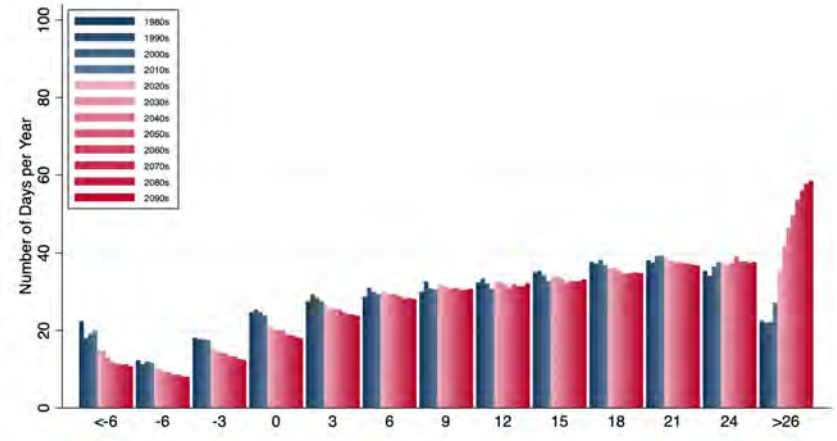
Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE II: DISTRIBUTION OF TEMPERATURE DAYS BY BIN OVER TIME

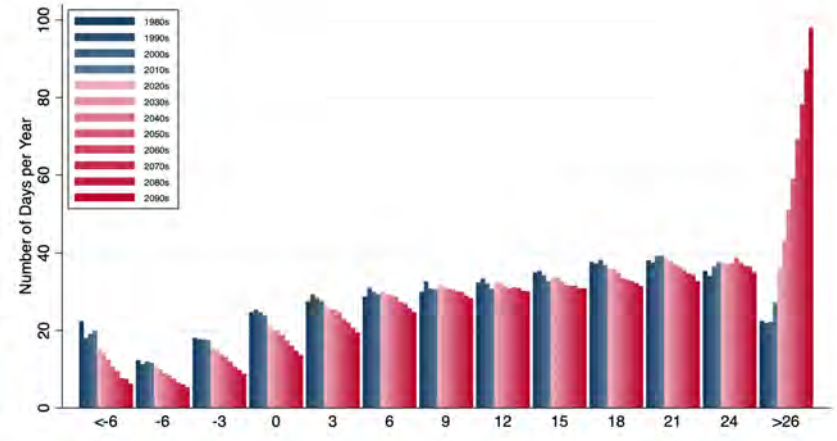
(a) RCP 2.6



(b) RCP 4.5



(c) RCP 8.5

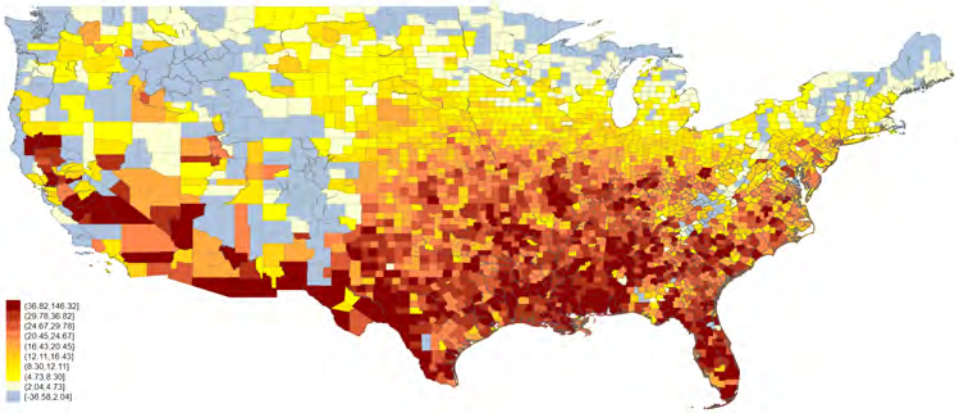


The figure shows decadal U.S. long-run temperature projections on the basis of big data generated by Hsiang et al. (2017) for the 1980s to 2090s. The underlying data is based on 44 climate models. Shown are projections under three different Representative Concentration Pathways (RCPs) used to describe scenarios of greenhouse gas (GHG) emissions and atmospheric concentrations: a stringent mitigation scenario (Panel A, RCP 2.6), an intermediate scenario (Panel B, RCP 4.5), and a scenario with very high GHG emissions (Panel C, RCP 8.5). Each bar shows the average annual number of days for respective 3°C bins (x-axis) for a range of decades (color-coded bar), averaged across U.S. counties.

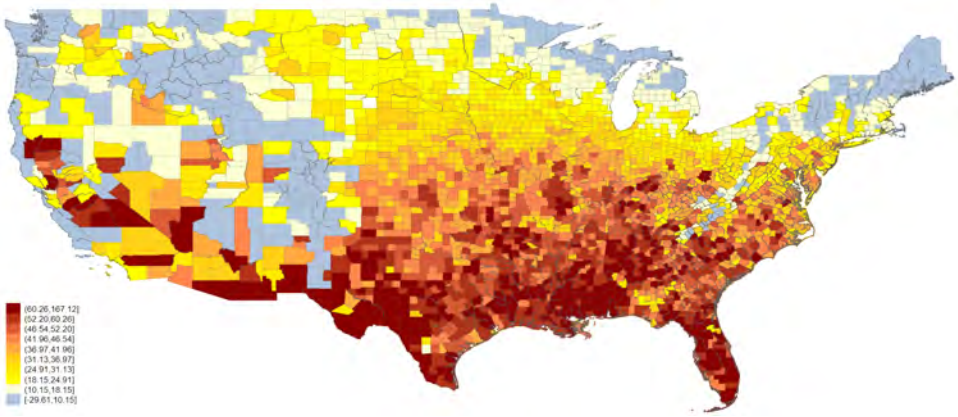
Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE III: PROJECTED CHANGES IN THE NUMBER OF DAYS ABOVE 26°C BETWEEN THE 1980S AND THE 2090S

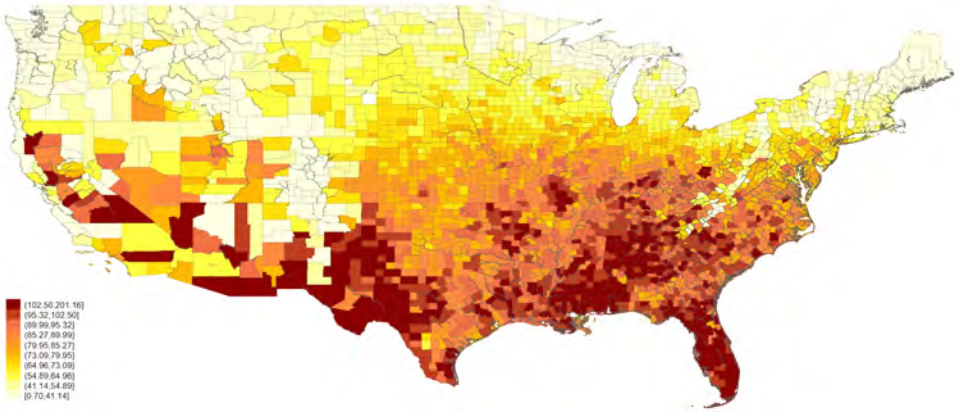
(a) RCP26



(b) RCP45



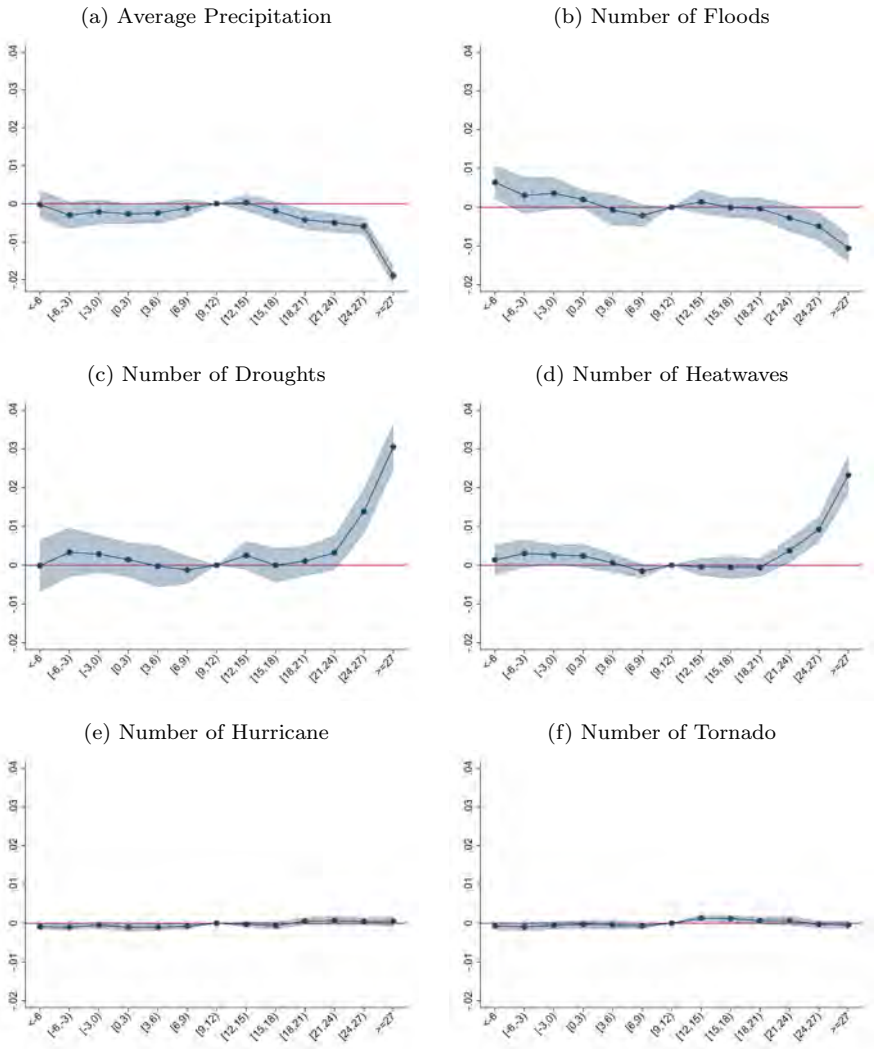
(c) RCP85



The figure shows long-run temperature projections by U.S. county for the contiguous 48 states on the basis of big data generated by Hsiang et al. (2017) for the 1980s to 2090s. The underlying data is based on 44 climate models. Shown is the county-level change between the average projected number of days above 26°C in the 2090s and the average number of days above 26°C in the 1980s for a stringent mitigation scenario (Panel A), an intermediate scenario (Panel B), and a scenario with very high GHG emissions (Panel C).

Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE IV: TEMPERATURE, PRECIPITATION, AND EXTREME WEATHER EVENTS

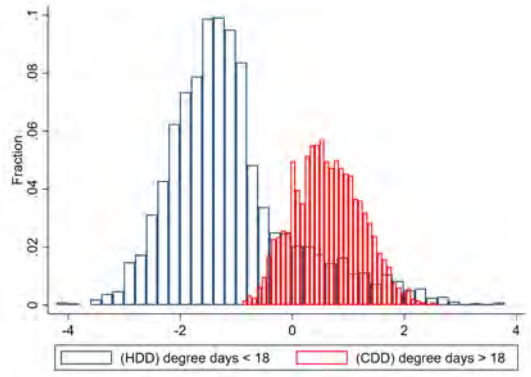


The figure shows the point estimates and the 95% confidence interval of county-year level regressions of controls that capture the number of days in various temperature bins (x-axis), county fixed effects, and year fixed effects on average precipitation (Panel A), number of floods (Panel B), number of droughts (Panel C), number of heatwaves (Panel D), number of hurricanes (Panel E), and number of tornados (Panel F). Standard errors are clustered at the state level. Event data is from SHELDUS and temperature and precipitation data from the PRISM Climate Group (we use the cleaned version of that data provided on Wolfram Schlenker's homepage).

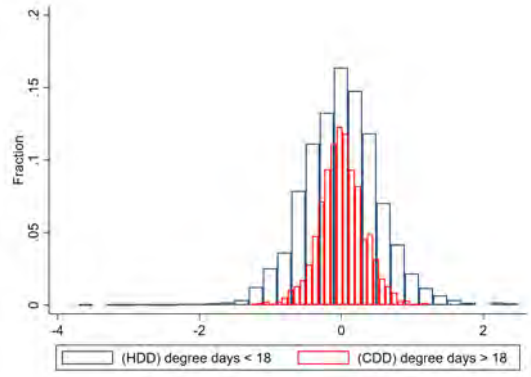
Electronic copy available at: https://ssrn.com/abstract=4526111

Figure V: Distribution of the long-run changes in degree days above and below 18°C

(a) Raw changes



(b) Residual changes after removing state FE

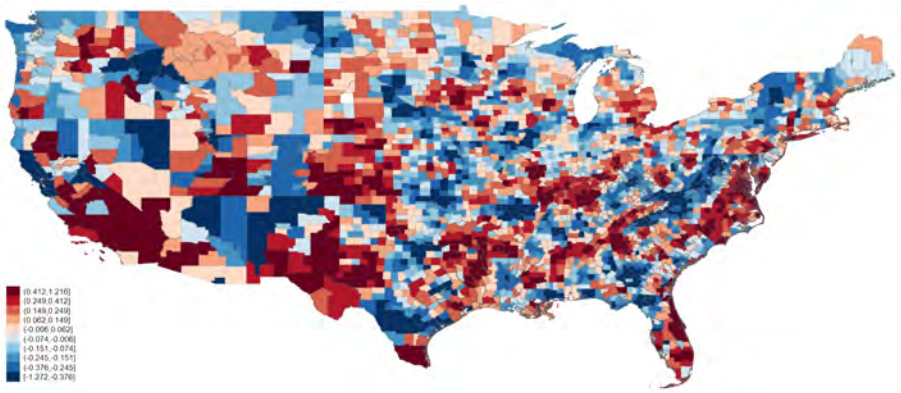


The figure shows the distribution of changes in Heating Degree Days (HDD; blue) and Cooling Degree Days (CDDs; red) from the 1980s to the 2010s at the county level. A daily Cooling Degree Day (CDD) is the difference in degrees between the average daily temperature and 18°C conditional on the average daily temperature being above 18°C, and a daily Heating Degree Day (HDD) is the difference in degrees between 18°C and the average daily temperature conditional on the average daily temperature being below 18°C, see Heutel et al. (2021) or Zivin and Kahn (2016). For each county, average daily HDDs and CDDs are summed by year, yearly HDDs and CDDs are then averaged over the 2010s and 1980s, respectively, from which the long-run difference is calculated. Underlying data is from the PRISM Climate Group (we use the cleaned version of that data provided on Wolfram Schlenker's homepage). Panel A shows raw data, and Panel B shows the distribution after removing state fixed effects.

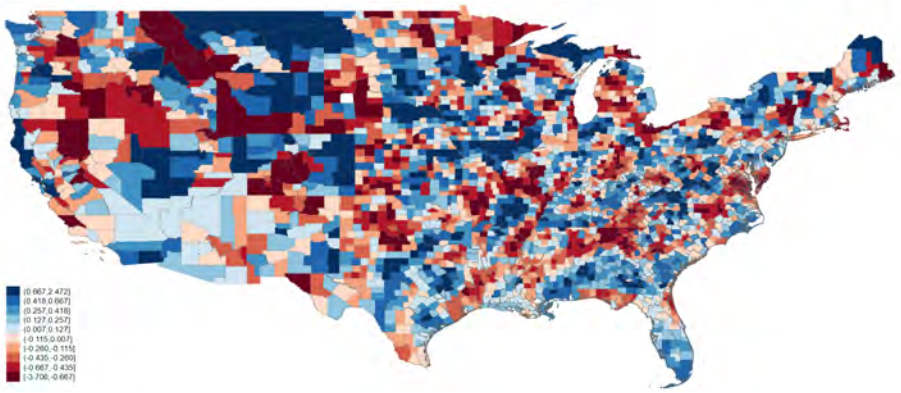
Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE VI: GEOGRAPHIC DISTRIBUTION OF LONG-RUN CHANGES IN DEGREE DAYS ABOVE AND BELOW 18°C

(a) Δ (degree days > 18°C)



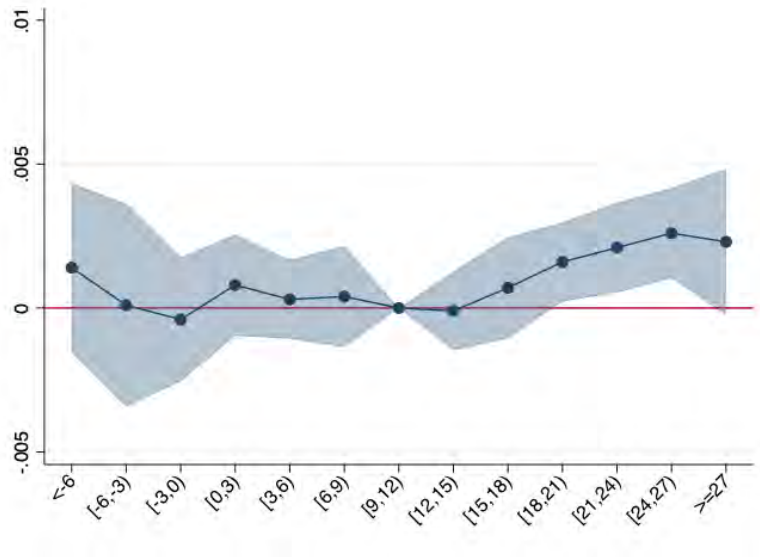
(b) Δ (degree days < 18°C)



The figure shows changes in Cooling Degree Days (CDDs, Panel A) and Heating Degree Days (HDDs, Panel B) between the 1980s and the 2010s by county. A daily Cooling Degree Day (CDD) is the difference in degrees between the average daily temperature and 18°C conditional on the average daily temperature being above 18°C, and a daily Heating Degree Day (HDD) is the difference in degrees between 18°C and the average daily temperature conditional on the average daily temperature being below 18°C, see Heutel et al. (2021) or Zivin and Kahn (2016). For each county, average daily HDDs and CDDs are summed by year, yearly HDDs and CDDs are then averaged over the 2010s and 1980s, respectively, from which the long-run difference is calculated. Underlying data is from the PRISM Climate Group (we use the cleaned version of that data provided on Wolfram Schlenker's homepage). Red indicates counties that have become warmer, i.e., that experienced an increase in CDDs (Panel A) or a decrease in HDDs (Panel B) relative to state-level changes.

Electronic copy available at: https://ssrn.com/abstract=4526111

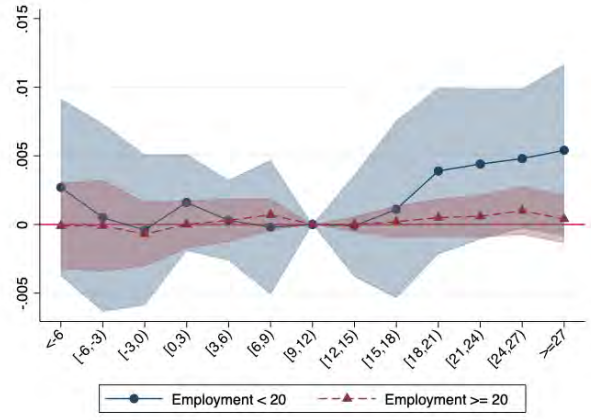
FIGURE VII: SHORT-RUN EFFECT OF TEMPERATURE ON ENERGY COSTS



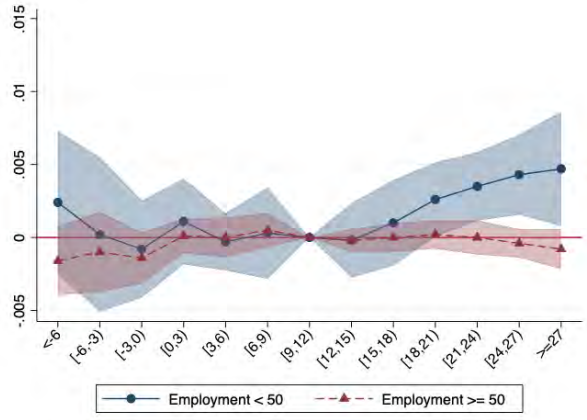
The figure shows the point estimates and the 95% confidence intervals when using the panel data approach described in Equation (1) to estimate the short-run effect of temperature on energy costs divided by total value of shipments. The analysis is at the plant-year level and the sample period comprises 1977-2018. The coefficients shown along the x-axis represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. Control variables include average zip code-year level precipitation, as well as number of hurricanes and number of tornadoes at the county-year level. Plant, industry-year, and state-year fixed effects are also included. Underlying regressions are estimated using ASM sample weights.

FIGURE VIII: HETEROGENEOUS SHORT-RUN EFFECTS OF TEMPERATURE ON ENERGY COSTS

(a) Split at Employment=20



(b) Split at Employment=50

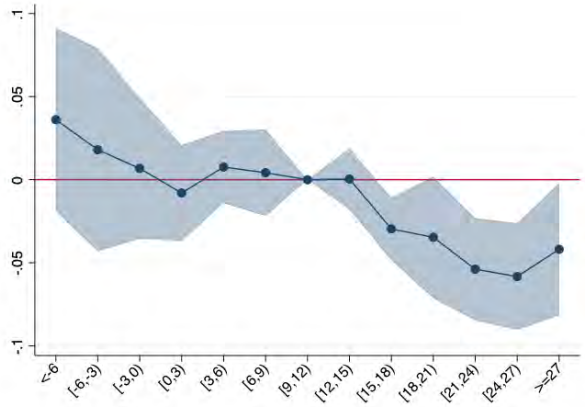


The figure shows point estimates and 95% confidence intervals when estimating the specification used in Figure VII by plant size. In Panel A, the sample is split into plants with strictly fewer than 20 employees (blue) and more than 20 employees (red). In Panel B, the size cut-off is 50 employees. See Figure VII and Table II for further details.

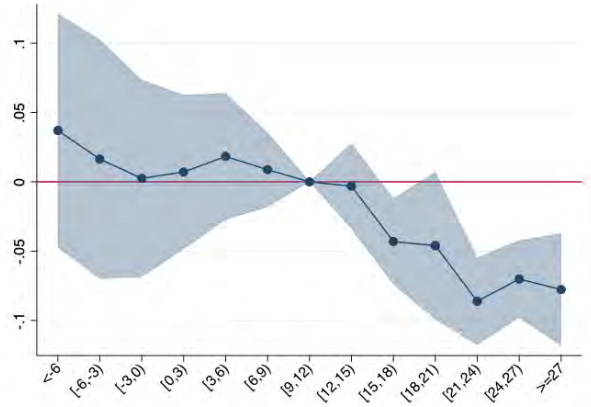
Electronic copy available at: https://ssrn.com/abstract=4526111

Figure IX: Short-run effect of temperature on productivity

(a) Log(TFP)



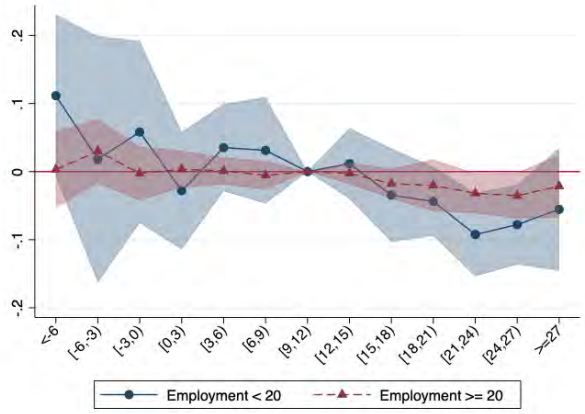
(b) Log(Value Added/Total Working Hours)



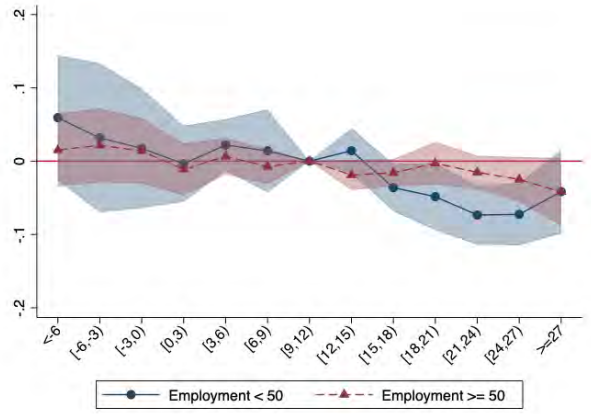
The figure shows the point estimates and the 95% confidence intervals when using the panel data approach described in Equation (1) to estimate the short-run effects of temperature on the natural logarithm of total factor productivity (TFP; Panel A) and the natural logarithm of value added over total working hours (Panel B). The analysis is at the plant-year level and the sample period comprises 1977-2018. The coefficients shown along the x-axis represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. Control variables include average zip code-year level precipitation, as well as number of hurricanes and number of tornadoes at the county-year level. Plant, industry-year, and state-year fixed effects are also included. Underlying regressions are estimated using ASM sample weights.

Figure X: Heterogeneous short-run effects of temperature on Log(TFP)

(a) Split at employment=20



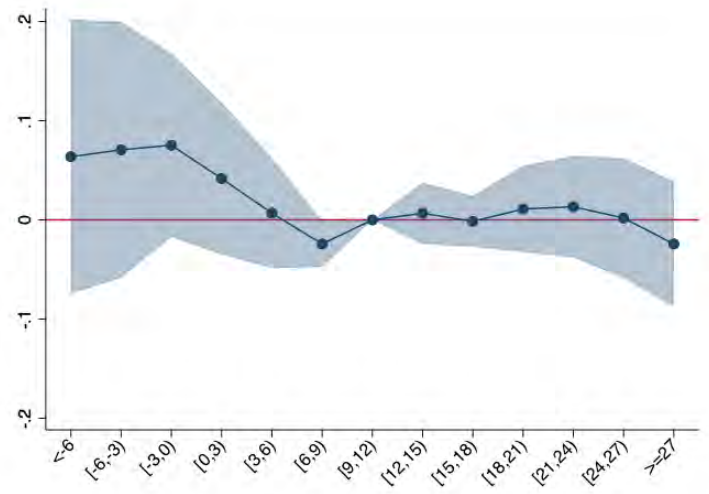
(b) Split at employment=50



The figure shows point estimates and 95% confidence intervals when estimating the specification used in Figure IX Panel A (natural logarithm of TFP) by plant size. In Panel A, the sample is split into plants with strictly fewer than 20 employees (blue) and more than 20 employees (red). In Panel B, the size cut-off is 50 employees. See Figure IX and Table III for further details.

Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE XI: SHORT-RUN EFFECT OF TEMPERATURE ON TOTAL EMPLOYEE-HOURS

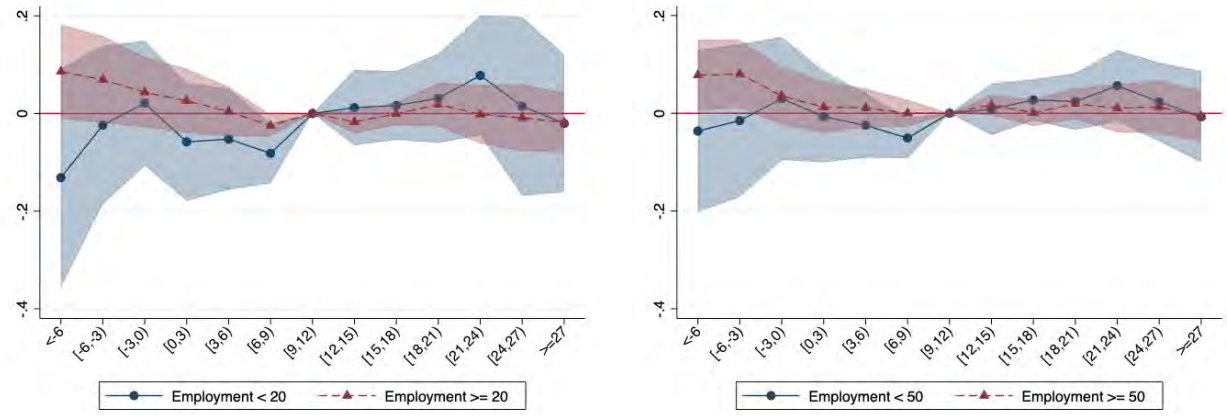


The figure shows the point estimates and the 95% confidence intervals when using the panel data approach described in Equation (1) to estimate the short-run effect of temperature on the natural logarithm of total hours worked. The analysis is at the plant-year level and the sample period comprises 1977-2018. The coefficients shown along the x-axis represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. Control variables include average zip code-year level precipitation, as well as number of hurricanes and number of tornadoes at the county-year level. Plant, industry-year, and state-year fixed effects are also included. Underlying regressions are estimated using ASM sample weights.

FIGURE XII: HETEROGENEOUS SHORT-RUN EFFECTS OF TEMPERATURE ON TOTAL EMPLOYEE-HOURS

(a) Employment < 20

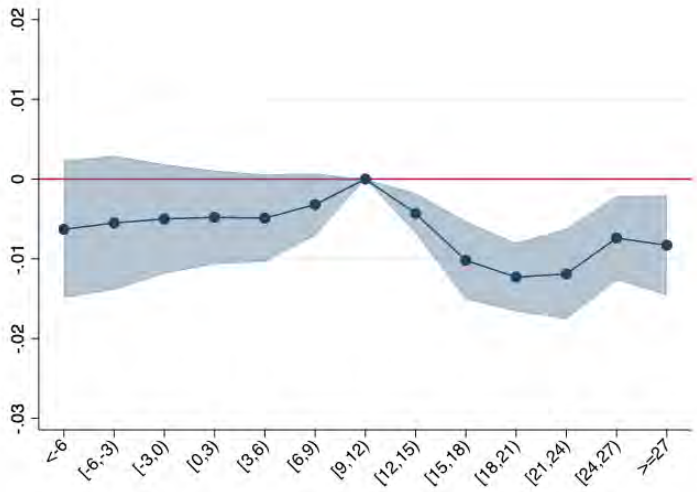
(b) Employment < 50



The figure shows point estimates and 95% confidence intervals when estimating the specification used in Figure XI by plant size. In Panel A, the sample is split into plants with strictly fewer than 20 employees (blue) and more than 20 employees (red). In Panel B, the size cut-off is 50 employees. See Figure XI and Table IV for further details.

Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE XIII: SHORT-RUN EFFECT OF TEMPERATURE ON ENTRY

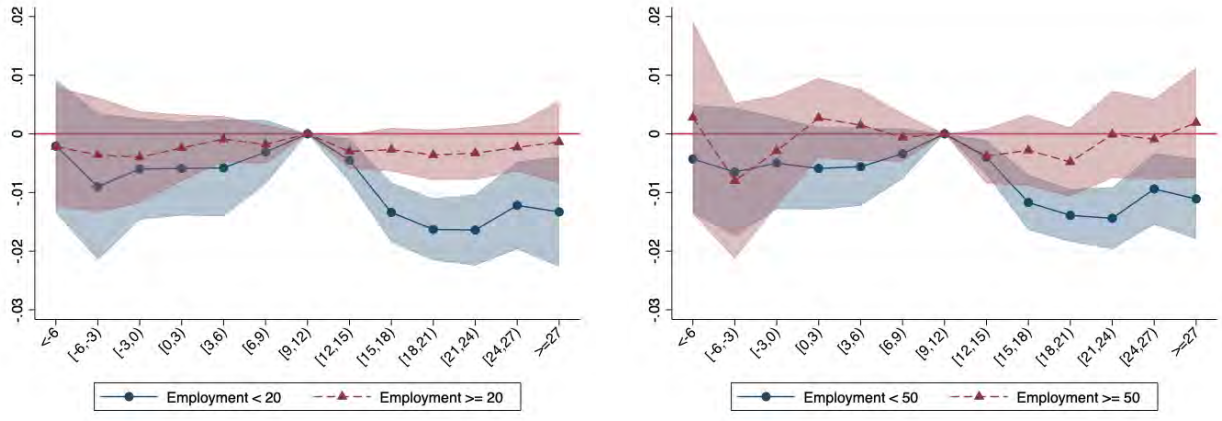


The figure shows the point estimates and the 95% confidence intervals when using the panel data approach described in Equation (1) to estimate the short-run effect of temperature on entry. The analysis is at the plant-year level and the sample period comprises 1977-2018. The coefficients shown along the x-axis represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. Control variables include average zip code-year level precipitation, as well as number of hurricanes and number of tornadoes at the county-year level. Plant, industry-year, and state-year fixed effects are also included.

FIGURE XIV: HETEROGENEOUS SHORT-RUN EFFECTS OF TEMPERATURE ON ENTRY

(a) Split at employment=20

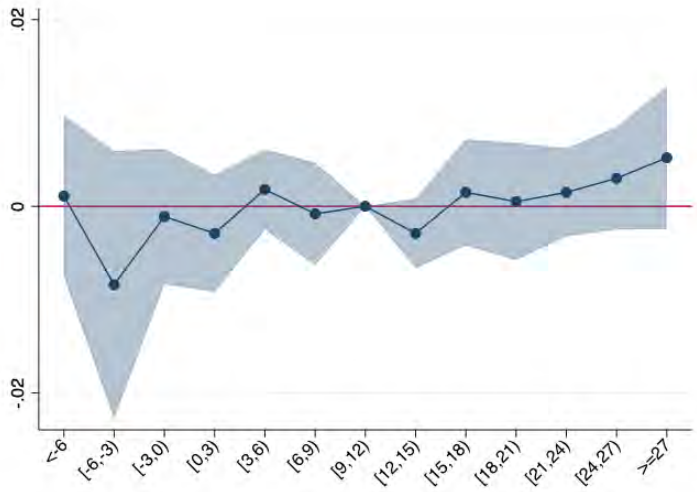
(b) Split at employment=50



The figure shows point estimates and 95% confidence intervals when estimating the specification used in Figure XIII by plant size. In Panel A, the sample is split into plants with strictly fewer than 20 employees (blue) and more than 20 employees (red). In Panel B, the size cut-off is 50 employees. See Figure XIII and Table V for further details.

Electronic copy available at: https://ssrn.com/abstract=4526111

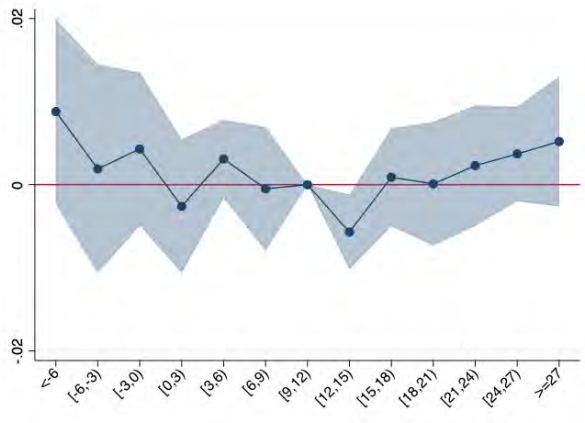
FIGURE XV: SHORT-RUN EFFECT OF TEMPERATURE ON EXIT



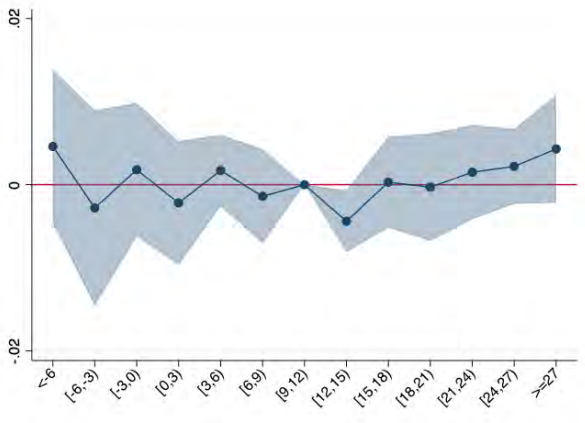
The figure shows the point estimates and the 95% confidence intervals when using the panel data approach described in Equation (1) to estimate the short-run effect of temperature on exit. The analysis is at the plant-year level and the sample period comprises 1977-2018. The coefficients shown along the x-axis represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. Control variables include average zip code-year level precipitation, as well as number of hurricanes and number of tornadoes at the county-year level. Plant, industry-year, and state-year fixed effects are also included.

FIGURE XVI: HETEROGENEOUS SHORT-RUN EFFECTS OF TEMPERATURE ON EXIT

(a) Employment < 20



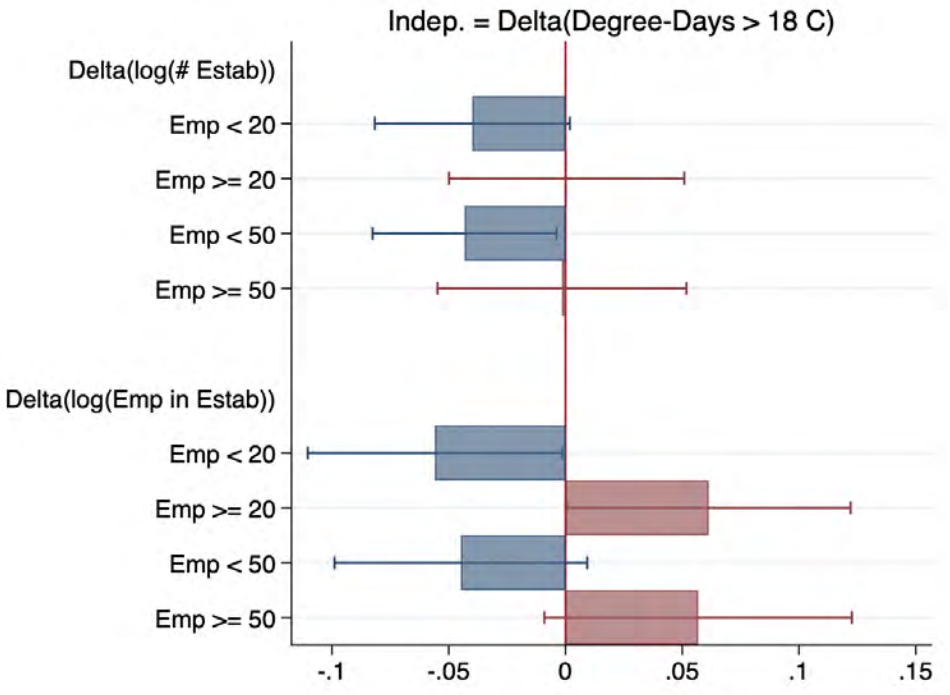
(b) Employment < 50



The figure shows point estimates and 95% confidence intervals when estimating the specification used in Figure XV by plant size. In Panel A, the sample is based on plants strictly fewer than 20 employees. In Panel B, the size cut-off is 50 employees. See Figure XV and Table V for further details.

Electronic copy available at: https://ssrn.com/abstract=4526111

FIGURE XVII: HETEROGENEOUS LONG-RUN EFFECTS OF TEMPERATURE ON NUMBER OF PLANTS AND EMPLOYMENT



The figure illustrates the results of Table VI by showing the point estimates and the 95% confidence intervals when using the county-level long-run approach described in Equation (2) to estimate the long-run effects of temperature on the number of establishments and number of employees across plant sizes, defined as strictly fewer than 20 (50) employees (blue) and more than 20 (50) employees (red). The top panel shows results for the change in the natural logarithm of the average yearly number of establishments between the 2010s and the 1980s. Similarly, the bottom panel shows results for the change in the natural logarithm of the average yearly number of employees reported by establishments. The shown coefficient of interest is the change in average CDDs between the 2010s and the 1980s at the county level. Included throughout are state fixed effects and controls for long-run changes in average precipitation, percentage of population that attended at least one year of college in 1980, log-transformed population in 1980, log-transformed income per capita in 1980, changes in exposure to the China shock between 1990 and 2007 (Autor et al. 2013), and changes in occurrences of hurricanes and tornados between the 2010s and the 1980s. Standard errors are clustered at the state level.

Electronic copy available at: https://ssrn.com/abstract=4526111

VIII Tables

Table I: Summary Statistics

Variables	N	Mean	Sd
Panel A: ASM & CMF Sample			
Energy/Total Value of Shipments	1922000	0.022	0.0291
Log(TFP)	1922000	1.85	0.56
Log(Value-Added / Total Hours Worked)	1922000	3.472	0.912
Log(Total Hours Worked)	1922000	5.193	1.395
T < -6 ˚C	1922000	15.34	19.09
-6 ˚C ≤ T < -3 ˚C	1922000	10.92	9.622
-3 ˚C ≤ T < 0 ˚C	1922000	16.48	12.41
0 ˚C ≤ T < 3 ˚C	1922000	22.57	14.14
3 ˚C ≤ T < 6 ˚C	1922000	26.47	13.69
6 ˚C ≤ T < 9 ˚C	1922000	29.4	12.76
12 ˚C ≤ T < 15 ˚C	1922000	35.66	14.14
15 ˚C ≤ T < 18 ˚C	1922000	38.51	15.82
18 ˚C ≤ T < 21 ˚C	1922000	41.95	14.39
21 ˚C ≤ T < 24 ˚C	1922000	42.36	15.01
24 ˚C ≤ T < 27 ˚C	1922000	32.96	22.25
T ≥ 27 ˚C	1922000	20.02	30.94
Panel B: LBD Sample			
Exit	13590000	0.0754	0.264
Entry	13590000	0.0727	0.26
T < -6 ˚C	13590000	14.21	18.59
-6 ˚C ≤ T < -3 ˚C	13590000	10.47	9.739
-3 ˚C ≤ T < 0 ˚C	13590000	15.77	12.67
0 ˚C ≤ T < 3 ˚C	13590000	21.65	14.83
3 ˚C ≤ T < 6 ˚C	13590000	25.68	14.97
6 ˚C ≤ T < 9 ˚C	13590000	28.87	14.14
12 ˚C ≤ T < 15 ˚C	13590000	36.95	16.12
15 ˚C ≤ T < 18 ˚C	13590000	40.04	17.95
18 ˚C ≤ T < 21 ˚C	13590000	42.79	15.84
21 ˚C ≤ T < 24 ˚C	13590000	42.74	16.59
24 ˚C ≤ T < 27 ˚C	13590000	32.28	23.24
T ≥ 27 ˚C	13590000	20.81	33.02
Panel C: LBD - Long-run difference between the 2010s and the 1980s			
Δ Degree-Days > 18 ˚C / 100	2800	1.197	1.058
Δ Degree-Days < 18 ˚C / 100	2800	-2.399	1.536
Δ Log(# Estab.)	2800	-0.0087	0.389
Δ Log(Emp.)	2800	-0.241	0.721
Δ Log(Avg. Size of Estab.)	2800	-0.235	0.61
Δ Fraction of Emp. in Top 5 Largest Estab.	2800	-0.0111	0.112
Δ Log(HHI_Emp.)	2800	-0.0349	0.563
Δ Log(# Estab. of Size < 20)	2800	0.042	0.438
Δ Log(# Estab. of Size ≥ 20)	2800	-0.111	0.589
Δ Log(Emp. in Estab. of Size < 20)	2800	-0.0045	0.495
Δ Log(Emp. in Estab. of Size ≥ 20)	2800	-0.289	0.757
Δ Log(# Estab. of Size < 50)	2500	0.0355	0.394
Δ Log(# Estab. of Size ≥ 50)	2500	-0.198	0.653
Δ Log(Emp. in Estab. of Size < 50)	2500	-0.0072	0.501
Δ Log(Emp. in Estab. of Size ≥ 50)	2500	-0.312	0.752

Notes: The table shows summary statistics for the ASM and CMF sample at the plant-year level (Panel A), the LBD sample at the plant-year level (Panel B), and long-run differences (average in the 2010s minus average in the 1980s) for the LBD sample at the county level (Panel C). Variable definitions are in Appendix Table A.1.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table II: Short-run effect of temperature on energy costs

Dep. Var.	Energy Costs/TVS			
	(1)	(2)	(3)	(4)
T < -6 ˚C	0.0004	0.0008	0.0014	0.0014
	(0.0007)	(0.0006)	(0.0015)	(0.0015)
-6 ˚C ≤ T < -3 ˚C	0.0009	0.0008	0.0001	0.0001
	(0.0009)	(0.0009)	(0.0018)	(0.0018)
-3 ˚C ≤ T < 0 ˚C	0.0002	0	-0.0004	-0.0004
	(0.0006)	(0.0006)	(0.0011)	(0.0011)
0 ˚C ≤ T < 3 ˚C	0.0008	0.0008	0.0008	0.0008
	(0.0005)	(0.0005)	(0.0009)	(0.0009)
3 ˚C ≤ T < 6 ˚C	0.0006	0.0003	0.0003	0.0003
	(0.0006)	(0.0006)	(0.0007)	(0.0007)
6 ˚C ≤ T < 9 ˚C	0.0004	0.0003	0.0005	0.0004
	(0.0007)	(0.0007)	(0.0009)	(0.0009)
12 ˚C ≤ T < 15 ˚C	0.0008	0.0009**	-0.0001	-0.0001
	(0.0005)	(0.0004)	(0.0007)	(0.0007)
15 ˚C ≤ T < 18 ˚C	0.0013***	0.0012**	0.0007	0.0007
	(0.0005)	(0.0004)	(0.0009)	(0.0009)
18 ˚C ≤ T < 21 ˚C	0.0015***	0.0014***	0.0017**	0.0016**
	(0.0005)	(0.0005)	(0.0007)	(0.0007)
21 ˚C ≤ T < 24 ˚C	0.0017***	0.0015***	0.0021***	0.0021***
	(0.0005)	(0.0005)	(0.0008)	(0.0008)
24 ˚C ≤ T < 27 ˚C	0.0013***	0.0013***	0.0026***	0.0026***
	(0.0004)	(0.0004)	(0.0008)	(0.0008)
T ≥ 27 ˚C	0.0007	0.0008	0.0023*	0.0023*
	(0.0006)	(0.0005)	(0.0013)	(0.0013)
Obs	1922000	1922000	1922000	1922000
R-squared	0.786	0.793	0.795	0.795
Establishment FE	yes	yes	yes	yes
Year FE	yes			
NAICS4-Year FE		yes	yes	yes
State-year FE			yes	yes
Extreme weather controls				yes
Se. cluster level	State	State	State	State
Sample period	1977-2018	1977-2018	1977-2018	1977-2018

Notes: The table uses the panel data approach described in Equation (1) to estimate the short-run effect of temperature on energy costs divided by total value of shipments. The analysis is at the plant-year level and the sample period comprises 1977-2018. The shown coefficients of interest represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9˚C,12˚C) is used as the reference bin and therefore omitted. All specifications control for average zip code-year precipitation and include establishment fixed effects. Further fixed effects are included as indicated. Extreme weather controls include number of hurricanes and number of tornadoes, both of which are measured at the county-year level. Regressions are estimated using ASM sample weights. Reported standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table III: Short-run effect of temperature on productivity

Dep. Var.	Log(TFP)				Log(Value-Added/Total Hours Worked)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T < -6 ˚C	0.0044	-0.0024	0.0356	0.0361	0.0129	0.0068	0.0365	0.0371
	(0.0125)	(0.0107)	(0.0280)	(0.0281)	(0.0189)	(0.0176)	(0.0430)	(0.0430)
-6 ˚C ≤ T < -3 ˚C	-0.0248*	-0.0147	0.018	0.018	-0.0299	-0.0237	0.0168	0.0164
	(0.0136)	(0.0136)	(0.0311)	(0.0312)	(0.0202)	(0.0212)	(0.0441)	(0.0441)
-3 ˚C ≤ T < 0 ˚C	0.0004	-0.0029	0.0065	0.0068	0.0024	-0.0025	0.0022	0.0025
	(0.0147)	(0.0133)	(0.0214)	(0.0216)	(0.0197)	(0.0192)	(0.0361)	(0.0363)
0 ˚C ≤ T < 3 ˚C	0.0044	-0.0033	-0.008	-0.008	0.0057	0.0027	0.0072	0.0071
	(0.0105)	(0.0090)	(0.0146)	(0.0147)	(0.0133)	(0.0128)	(0.0285)	(0.0284)
3 ˚C ≤ T < 6 ˚C	0.0019	0.0012	0.0077	0.0076	0.0035	0.0012	0.0183	0.0183
	(0.0117)	(0.0096)	(0.0111)	(0.0111)	(0.0148)	(0.0137)	(0.0235)	(0.0233)
6 ˚C ≤ T < 9 ˚C	-0.0062	-0.0107	0.0045	0.0042	-0.0064	-0.0081	0.0089	0.0088
	(0.0103)	(0.0104)	(0.0132)	(0.0133)	(0.0132)	(0.0126)	(0.0138)	(0.0138)
12 ˚C ≤ T < 15 ˚C	-0.0061	-0.0073	0.0011	0.0004	0.0023	-0.0007	-0.0021	-0.0031
	(0.0091)	(0.0086)	(0.0094)	(0.0094)	(0.0141)	(0.0129)	(0.0159)	(0.0158)
15 ˚C ≤ T < 18 ˚C	0.0056	-0.0012	-0.0292***	-0.0296***	-0.0117	-0.0131	-0.0426**	-0.0430***
	(0.0129)	(0.0109)	(0.0097)	(0.0096)	(0.0122)	(0.0121)	(0.0159)	(0.0158)
18 ˚C ≤ T < 21 ˚C	0	0.0004	-0.0343*	-0.0347*	-0.0024	-0.0027	-0.0454	-0.0460*
	(0.0105)	(0.0093)	(0.0188)	(0.0187)	(0.0134)	(0.0122)	(0.0272)	(0.0272)
21 ˚C ≤ T < 24 ˚C	-0.0022	-0.0086	-0.0531***	-0.0539***	-0.0275**	-0.0310**	-0.0851***	-0.0861***
	(0.0119)	(0.0097)	(0.0158)	(0.0157)	(0.0126)	(0.0141)	(0.0161)	(0.0161)
24 ˚C ≤ T < 27 ˚C	-0.0042	-0.0101	-0.0573***	-0.0583***	-0.0230*	-0.0246*	-0.0689***	-0.0701***
	(0.0103)	(0.0097)	(0.0165)	(0.0164)	(0.0128)	(0.0132)	(0.0141)	(0.0142)
T ≥ 27 ˚C	0.0106	0.0068	-0.0413**	-0.0419**	-0.0023	-0.0048	-0.0769***	-0.0777***
	(0.0089)	(0.0088)	(0.0203)	(0.0203)	(0.0125)	(0.0121)	(0.0205)	(0.0207)
Obs	1922000	1922000	1922000	1922000	1922000	1922000	1922000	1922000
R-squared	0.771	0.785	0.787	0.787	0.777	0.781	0.782	0.782
Establishment FE	yes	yes	yes	yes	yes	yes	yes	yes
Year FE	yes				yes			
NAICS4-Year FE		yes	yes	yes		yes	yes	yes
State-year FE			yes	yes			yes	yes
Extreme Weather controls				yes				yes
Se. cluster level	State	State	State	State	State	State	State	State
Sample period	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018

Notes: The table uses the panel data approach described in Equation (1) to estimate the short-run effects of temperature on the natural logarithm of total factor productivity (TFP; Columns (1)-(4)) and the natural logarithm of value added divided by total number of hours worked (Columns (5)-(8)). The analysis is at the plant-year level and the sample period comprises 1977-2018. The shown coefficients of interest represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9˚C,12˚C) is used as the reference bin and therefore omitted. All specifications control for average zip code-year precipitation and include establishment fixed effects. Further fixed effects are included as indicated. Extreme weather controls include number of hurricanes and number of tornadoes, both of which are measured at the county-year level. Regressions are estimated using ASM sample weights. Reported standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table IV: Short-run effect of temperature on employment

Dep. Var.	Log(Total Hours Worked)			
	(1)	(2)	(3)	(4)
T < -6 °C	0.0487**	0.0484**	0.0643	0.0637
	(0.0186)	(0.0181)	(0.0704)	(0.0705)
-6 °C ≤ T < -3 °C	0.0321	0.0382	0.0709	0.0706
	(0.0375)	(0.0339)	(0.0660)	(0.0658)
-3 °C ≤ T < 0 °C	0.0497**	0.0447**	0.0756	0.0753
	(0.0216)	(0.0214)	(0.0472)	(0.0472)
0 °C ≤ T < 3 °C	0.0342	0.0303	0.0418	0.0417
	(0.0207)	(0.0199)	(0.0392)	(0.0391)
3 °C ≤ T < 6 °C	0.0025	0.006	0.0066	0.0068
	(0.0163)	(0.0150)	(0.0283)	(0.0283)
6 °C ≤ T < 9 °C	0.004	0.0033	-0.0247**	-0.0243**
	(0.0135)	(0.0119)	(0.0118)	(0.0118)
12 °C ≤ T < 15 °C	-0.0008	-0.0022	0.0057	0.0066
	(0.0232)	(0.0191)	(0.0158)	(0.0158)
15 °C ≤ T < 18 °C	-0.0249	-0.022	-0.0021	-0.0015
	(0.0166)	(0.0148)	(0.0132)	(0.0131)
18 °C ≤ T < 21 °C	-0.0138	-0.0166	0.0107	0.011
	(0.0196)	(0.0162)	(0.0224)	(0.0222)
21 °C ≤ T < 24 °C	-0.0199	-0.019	0.0122	0.0132
	(0.0199)	(0.0186)	(0.0263)	(0.0261)
24 °C ≤ T < 27 °C	-0.0191	-0.0147	0.0008	0.0019
	(0.0136)	(0.0126)	(0.0312)	(0.0309)
T ≥ 27 °C	-0.0092	-0.0124	-0.025	-0.0243
	(0.0241)	(0.0230)	(0.0322)	(0.0322)
Obs	1922000	1922000	1922000	1922000
R-squared	0.923	0.925	0.925	0.925
Establishment FE	yes	yes	yes	yes
Year FE	yes			
NAICS4-Year FE		yes	yes	yes
State-year FE			yes	yes
Extreme Weather controls				yes
Se. cluster level	State	State	State	State
Sample period	1977-2018	1977-2018	1977-2018	1977-2018

Notes: The table uses the panel data approach described in Equation (1) to estimate the short-run effect of temperature on the natural logarithm of total number of hours worked. The analysis is at the plant-year level and the sample period comprises 1977-2018. The shown coefficients of interest represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9°C,12°C) is used as the reference bin and therefore omitted. All specifications control for average zip code-year precipitation and include establishment fixed effects. Further fixed effects are included as indicated. Extreme weather controls include number of hurricanes and number of tornadoes, both of which are measured at the county-year level. Regressions are estimated using ASM sample weights. Reported standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table V: Short-run effects of temperature on entry and exit

Dep. Var.	Entry				Exit			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T < -6 ˚C	-0.0031	-0.003	-0.0063	-0.0063	-0.0052	-0.0053	0.0011	0.0011
	(0.0076)	(0.0076)	(0.0044)	(0.0044)	(0.0042)	(0.0043)	(0.0043)	(0.0044)
-6 ˚C ≤ T < -3 ˚C	-0.0118	-0.0116	-0.0056	-0.0055	-0.0076	-0.0077	-0.0084	-0.0084
	(0.0094)	(6)	(0.0043)	(0.0043)	(0.0051)	(0.0051)	(0.0073)	(0.0073)
-3 ˚C ≤ T < 0 ˚C	-0.0045	-0.0045	-0.005	-0.005	-0.0024	-0.0024	-0.0011	-0.0011
	(0.0061)	(0.0062)	(0.0035)	(0.0035)	(0.0036)	(0.0036)	(0.0037)	(0.0037)
0 ˚C ≤ T < 3 ˚C	-0.0026	-0.0025	-0.0048	-0.0048	-0.0064**	-0.0064**	-0.0029	-0.0029
	(0.0055)	(0.0055)	(0.0030)	(0.0030)	(0.0030)	(0.0030)	(0.0032)	(0.0032)
3 ˚C ≤ T < 6 ˚C	-0.0001	-0.0002	-0.0049*	-0.0049*	-0.0002	-0.0001	0.0017	0.0018
	(0.0052)	(0.0051)	(0.0028)	(0.0028)	(0.0020)	(0.0020)	(0.0022)	(0.0022)
6 ˚C ≤ T < 9 ˚C	-0.0005	-0.0006	-0.003	-0.0032	-0.0007	-0.0006	-0.0009	-0.0008
	(0.0028)	(0.0027)	(0.0020)	(0.0020)	(0.0024)	(0.0024)	(0.0029)	(0.0028)
12 ˚C ≤ T < 15 ˚C	-0.0028	-0.0028	-0.0042***	-0.0043***	-0.0047	-0.0046	-0.0029	-0.0029
	(0.0020)	(0.0020)	(0.0013)	(0.0013)	(0.0032)	(0.0032)	(0.0019)	(0.0019)
15 ˚C ≤ T < 18 ˚C	-0.0059**	-0.0059**	-0.0102***	-0.0102***	-0.0017	-0.0017	0.0015	0.0015
	(0.0024)	(0.0025)	(0.0025)	(0.0025)	(0.0029)	(0.0029)	(0.0029)	(0.0029)
18 ˚C ≤ T < 21 ˚C	-0.0085***	-0.0085***	-0.0122***	-0.0123***	-0.0007	-0.0006	0.0005	0.0005
	(0.0026)	(0.0026)	(0.0022)	(0.0022)	(0.0027)	(0.0027)	(0.0032)	(0.0032)
21 ˚C ≤ T < 24 ˚C	-0.0068**	-0.0068**	-0.0118***	-0.0119***	0.0005	0.0006	0.0015	0.0015
	(0.0032)	(0.0032)	(0.0029)	(0.0029)	(0.0030)	(0.0030)	(0.0025)	(0.0024)
24 ˚C ≤ T < 27 ˚C	-0.0016	-0.0017	-0.0072***	-0.0074***	0.0021	0.0022	0.0029	0.003
	(0.0041)	(0.0041)	(0.0027)	(0.0027)	(0.0042)	(0.0042)	(0.0028)	(0.0028)
T ≥ 27 ˚C	-0.0035	-0.0035	-0.0083**	-0.0083**	0.0014	0.0015	0.0052	0.0052
	(0.0046)	(0.0046)	(0.0032)	(0.0032)	(0.0053)	(0.0052)	(0.0038)	(0.0039)
Obs	1922000	1922000	1922000	1922000	1922000	1922000	1922000	1922000
R-squared	0.021	0.021	0.189	0.189	0.016	0.016	0.19	0.19
Zipcode FE	yes	yes			yes	yes		
Establishment FE			yes	yes			yes	yes
NAICS4-Year FE	yes	yes	yes	yes	yes	yes	yes	yes
State-year FE	yes	yes	yes	yes	yes	yes	yes	yes
Extreme weather controls		yes		yes		yes		yes
Se. cluster level	State	State	State	State	State	State	State	State
Sample period	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018	1977-2018

Notes: The table uses the panel data approach described in Equation (1) to estimate the short-run effects of temperature on entry (Columns (1)-(4)) and exit (Columns (5)-(8)). Entry is an indicator variable set equal to one if a plant had zero employees at time t-1 and strictly more than zero employees at time t. Exit is an indicator variable set equal to one if a plant had strictly more than zero employees in year t and zero employees in year t+1. The analysis is at the plant-year level and the sample period comprises 1977-2018. The shown coefficients of interest represent the number of days in each respective temperature bin (β_b in Equation (1)) in a given year in a plant's zip code. The number of days in a temperature bin is divided by 100 for readibility and the temperature bin [9˚C,12˚C) is used as the reference bin and therefore omitted. All specifications control for average zip code-year precipitation. Zipcode fixed effects are include in columns (1),(2), (5), and (6), while establishment fixed effects are included in columns (3), (4), (7) and (8). Further fixed effects are included as indicated. Extreme weather controls include number of hurricanes and number of tornadoes, both of which are measured at the county-year level. Reported standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table VI: Heterogeneous long-run effects of temperature on number of plants and employment

Panel A: Heterogeneity by establishment size of less or more than 20 workers

Dep. Var.	Δ Log(# Estab. of Size < 20)		Δ Log(# Estab. of Size ≥ 20)		Δ Log(Emp. in Estab. of Size < 20)		Δ Log(Emp. in Estab. of Size ≥ 20)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Degree-Days > 18 °C / 100	-0.0494**	-0.0399*	-0.0073	0.0005	-0.0668**	-0.0559**	0.0573*	0.0613*
	(0.0209)	(0.0213)	(0.0271)	(0.0257)	(0.0261)	(0.0278)	(0.0316)	(0.0310)
Δ Degree-Days < 18 °C / 100	-0.0253*	-0.0196	0.0041	0.0115	-0.0259*	-0.0215	0.0310*	0.0366**
	(0.0149)	(0.0149)	(0.0133)	(0.0125)	(0.0140)	(0.0143)	(0.0165)	(0.0160)
Obs	2800	2800	2800	2800	2800	2800	2800	2800
R-squared	0.125	0.138	0.179	0.193	0.094	0.101	0.181	0.186
State FE	yes	yes	yes	yes	yes	yes	yes	yes
County controls		yes		yes		yes		yes
Se. cluster level	State	State	State	State	State	State	State	State

Panel B: Heterogeneity by establishment size of less or more than 50 workers

Dep. Var.	Δ Log(# Estab. of Size < 50)		Δ Log(# Estab. of Size ≥ 50)		Δ Log(Emp. in Estab. of Size < 50)		Δ Log(Emp. in Estab. of Size ≥ 50)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Degree-Days > 18 °C / 100	-0.0558***	-0.0432**	-0.0051	-0.0015	-0.0556**	-0.0448	0.0436	0.0568*
	(0.0186)	(0.0201)	(0.0294)	(0.0272)	(0.0247)	(0.0276)	(0.0344)	(0.0336)
Δ Degree-Days < 18 °C / 100	-0.0211**	-0.0165	-0.0137	-0.0048	-0.0007	0.0033	0.0123	0.0172
	(0.0102)	(0.0103)	(0.0153)	(0.0137)	(0.0102)	(0.0104)	(0.0194)	(0.0187)
Obs	2500	2500	2500	2500	2500	2500	2500	2500
R-squared	0.155	0.175	0.173	0.191	0.114	0.127	0.182	0.19
State FE	yes	yes	yes	yes	yes	yes	yes	yes
County controls		yes		yes		yes		yes
Se. cluster level	State	State	State	State	State	State	State	State

Notes: The table shows the main coefficients when estimating the county-level long-run specification described in Equation (2) to examine the long-run implications of temperature for number of establishments and establishment employees constructed for establishments with strictly fewer vs. more than 20 employees (Panel A) and for establishments with strictly fewer vs. more than 50 employees (Panel B). In Columns 1-4, the left-hand side variable is the change in the natural logarithm of the average yearly number of establishments between the 2010s and the 1980s. In Columns 5-8, the left-hand side variable is the change in the natural logarithm of the average yearly number of employees reported by establishments between the 2010s and the 1980s. Throughout, the reported coefficient of interest is the change in the average CDDs (HDDs) between the 2010s and the 1980s at the county level. State fixed effects and a control for the change in average precipitation between the 2010s and the 1980s at the county level are included throughout. Even-numbered columns additionally include county-level controls for preconditions (percentage of population that attended at least one year of college in 1980, log-transformed population in 1980, log-transformed income per capita in 1980), change in exposure to the China shock between 1990 and 2007 (Autor et al. 2013), and changes in occurrences of hurricanes and tornados between the 2010s and the 1980s. Standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Table VII: Long-run effects of temperature on number of plants, employment and plant size

Dep. Var.	Δ Log(# Estab.)		Δ Log(Emp.)		Δ Log(Avg. Size of Estab.)	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Degree-Days > 18 °C / 100	-0.0406*	-0.0311	0.0336	0.0441	0.0730***	0.0727***
	(0.0213)	(0.0213)	(0.0373)	(0.0354)	(0.0248)	(0.0242)
Δ Degree-Days < 18 °C / 100	-0.0169	-0.0097	0.0189	0.0264	0.0362**	0.0361**
	(0.0139)	(0.0135)	(0.0206)	(0.0197)	(0.0142)	(0.0139)
Obs	2800	2800	2800	2800	2800	2800
R-squared	0.19	0.211	0.192	0.199	0.114	0.116
State FE	yes	yes	yes	yes	yes	yes
County controls		yes		yes		yes
Se. cluster level	State	State	State	State	State	State

Notes: The table follows Table VI in showing the main coefficients when estimating the county-level long-run specification described in Equation (2) to examine the long-run implications of temperature on local manufacturing activity. In Columns (1)-(2), the left-hand side variable is the change in the natural logarithm of the average yearly number of establishments between the 2010s and the 1980s. In Columns (3)-(4), the left-hand side variable is change in the natural logarithm of the average yearly number of employees reported by establishments between the 2010s and the 1980s. In Columns (5)-(6), the left-hand side varible of interest is the change in the logged average establishment size between the 2010s and the 1980s. Throughout, the reported coefficient of interest is the change in the average CDDs (HDDs) between the 2010s and the 1980s at the county level. State fixed effects and a control for the change in average precipitation between the 2010s and the 1980s at the county level are included throughout. Even-numbered columns additionally include county-level controls for preconditions (percentage of population that attended at least one year of college in 1980, log-transformed population in 1980, log-transformed income per capita in 1980), change in exposure to the China shock between 1990 and 2007 (Autor et al. 2013), and changes in occurrences of hurricanes and tornados between the 2010s and the 1980s. Standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Table VIII: Long-run effect of temperature on industrial concentration

Dep. Var.	Δ Frac. of Emp. in Top 5 Largest Estab.		Δ Log(HHI_Emp.)	
	(1)	(2)	(3)	(4)
Δ Degree-Days > 18 °C / 100	0.0113**	0.0133***	0.0329	0.0483**
	(0.0047)	(0.0038)	(0.0222)	(0.0206)
Δ Degree-Days < 18 °C / 100	0.0068*	0.0048	0.0158	0.0081
	(0.0034)	(0.0029)	(0.0157)	(0.0143)
Obs	2800	2800	2800	2800
R-squared	0.09	0.12	0.073	0.1
State FE	yes	yes	yes	yes
County controls		yes		yes
Se. cluster level	State	State	State	State

Notes: The table follows Table VI in showing the main coefficients when estimating the county-level long-run specification described in Equation (2) to examine the long-run implications of temperature on industrial concentration. In Columns (1)-(2), the left-hand side variable is the change in the average yearly fraction of employees reported between the 2010s and the 1980s. In Columns (3)-(4), the left-hand side variable is the change in the natural logarithm of the average yearly Herfindahl-Hirschman Index between the 2010s and the 1980s, constructed on the basis of the number of employees reported by establishments. Throughout, the reported coefficient of interest is the change in the average CDDs (HDDs) between the 2010s the 1980s at the county level. State fixed effects and a control for the change in average precipitation between the 2010s and the 1980s at the county level are included throughout. Even-numbered columns additionally include county-level controls for preconditions (percentage of population that attended at least one year of college in 1980, log-transformed population in 1980, log-transformed income per capita in 1980), change in exposure to the China shock between 1990 and 2007 (Autor et al. 2013), and changes in occurrences of hurricanes and tornados between the 2010s and the 1980s. Standard errors are clustered at the state level. Significance is indicated at 1% (***), 5% (**), and 10% (*) level.

Electronic copy available at: https://ssrn.com/abstract=4526111

Appendix

Table A.1: Variable Definitions

Variable	Definition	Source
Panel A: ASM & CMF Sample		
Energy Costs / TVS	The ratio of energy costs to total value of shipments	
Log(TFP)	Log of total factor productivity	
Log(Value-Added/Total Hours Worked)	Log of the ratio of value-added to workers' total working hours	
Log(Total Hours Worked)	Log of workers' total working hours	
Panel B: LBD Sample		
Entry	An indicator of entry, where employment of the firm in year t-1 is zero and in year t is above zero	
Exit	An indicator of exit, where employment of the firm in year t is above-zero and in year t+1 is zero	
Panel C: LBD - Long-run difference between the 1980s and the 2010s		
Δ Degree-Days $>$ 18 °C	long-difference in the average degree days above 18 °C from 1980s to 2010s	
Δ Degree-Days $<$ 18 °C	long-difference in the average degree days below 18 °C from 1980s to 2010s	
Δ Log(# Establishments)	Long-difference in log of average total establishment from 1980s to 2010s	
Δ Log(Employment)	Long-difference in log of average total employment from 1980s to 2010s	
Δ Log(Avg. Size of Establishments)	long-difference in log of average employment size from 1980s to 2010s	
Δ Frac. of Emp. in Top 5 Largest Estab.	long-difference in the average fraction of employment from top 5 establishments from 1980s to 2010s	
Δ Log(HHI_Emp.)	long-difference in the average HHI of employment from 1980s to 2010s	
Δ Log(# Estab. of Size $<$ 20)	long-difference in the log of average number of establishments with $<$ 20 workers from 1980s to 2010s	
Δ Log(# Estab. of Size $\geq$ 20)	long-difference in the log of average number of establishments with $\geq$ 20 workers from 1980s to 2010s	
Δ Log(Emp. in Estab. of Size $<$ 20)	long-difference in the log of average employment of establishments with $<$ 20 workers from 1980s to 2010s	
Δ Log(Emp. in Estab. of Size $\geq$ 20)	long-difference in the log of average employment of establishments with $\geq$ 20 workers from 1980s to 2010s	
Δ Log(# Estab. of Size $<$ 50)	long-difference in the log of average number of establishments with $<$ 50 workers from 1980s to 2010s	
Δ Log(# Estab. of Size $\geq$ 50)	long-difference in the log of average number of establishments with $\geq$ 50 workers from 1980s to 2010s	
Δ Log(Emp. in Estab. of Size $<$ 50)	long-difference in the log of average employment of establishments with $<$ 50 workers from 1980s to 2010s	
Δ Log(Emp. in Estab. of Size $\geq$ 50)	long-difference in the log of average employment of establishments with $\geq$ 50 workers from 1980s to 2010s	
Panel D: Control Variables from Other Sources		
Avg. Precipitation	Average daily precipitation of a location-year	Database built by Wolfram Schlenker
# Events of Floods	Number of drought events in the county-year	SHEDULS from Arizona State University
# Events of Droughts	Number of drought events in the county-year	SHEDULS from Arizona State University
# Events of Heatwaves	Number of heatwave events in the county-year	SHEDULS from Arizona State University
# Events of Hurricane	Number of hurricane events in the county-year	SHEDULS from Arizona State University
# Events of Tornado	Number of tornado events in the county-year	SHEDULS from Arizona State University
Δ IPW	Changes in the exposure to the import shock from China from 1990 to 2007	Autor, Dorn, and Hanson (2013)
Perc. of college students	Percentage of 25-year old or above population finished at least one year of college	US Census
Log(Population)	Log of county population	Database built by Andrew Leuven
Log(Income pc)	Log of county per capita income	IPUSM

Electronic copy available at: https://ssrn.com/abstract=4526111

Table A.2: Balance Table

Panel A: County initial characteristics

Dep. Var.	log(pc income)	log(pop)	percentage of pop attended college	ΔIPW
	(1)	(2)	(3)	(4)
Δ Degree-days $>$ 18 / 100	-0.0481	0.121	-0.0292**	-0.410
	(0.0302)	(0.200)	(0.0145)	(0.474)
Obs	3,085	3,085	3,085	3,104
R-squared	0.024	0.110	0.025	0.013

Panel B: Long-run changes in the occurrences of natural hazards

Dep. Var.	Δ avg. precipitation	Δ flood	Δ drought	Δ heatwave	Δ hurricane	Δ tornado
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Degree-days $>$ 18 / 100	-0.143***	-0.318	1.507	0.369	0.301	-0.0374
	(0.0392)	(0.623)	(1.720)	(0.645)	(0.445)	(0.477)
Observations	3,105	3,105	3,105	3,105	3,105	3,105
R-squared	0.073	0.024	0.066	0.042	0.011	0.018

Notes: Outcome variables in Column (1)-(3) of Panel A are county characteristics observed in 1980 Census. Outcome variable in Column (4) of Panel A is the changes in exposure to China shock between 1991 and 2007, as is defined in Autor et al. (2013). The last two rows in Panel A report the mean and standard deviation of the corresponding outcome variable in each column. Outcome variables in Panel B is the difference between the occurrences of each natural disaster in the 1980s and the 2010s. The independent variables in both panels are the changes in the number of degree days above 18°C from the 1980s to the 2010s. The long run changes in degree days below 18°C are also controlled in all specifications.The independent variables are divided by 100 to make the report table easier to read. Standard errors are clustered at the state level. Significance level: *** p<0.01, ** p<0.05, * p<0.1.

Electronic copy available at: https://ssrn.com/abstract=4526111